# A COHERENT, POLARIMETRIC SAR SIMULATION OF FORESTS

## DESIGN DOCUMENT AND ALGORITHM SPECIFICATIONS

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## 1. Executive Summary

This document describes the software package **PolSARproSim**$^+$, designed to calculate simulated synthetic aperture radar (SAR) imagery of model forest stands. This software builds on PolSARproSim, originally delivered to the University of Rennes as part of the documentation for the contract "PolInSAR Coherent Scattering and Imaging Code Development", 2006 by Mark L. Williams. This document serves as a reference, and as a general guide to the construction and operation of **PolSARproSim**$^+$, a description of the algorithms coded and updates to the software. This document is not definitive but serves as a broad definition of overall capability, and contains many details of the implementation.

The document is divided into a number of sections. The introduction contains a statement of capability; a high-level description of **PolSARproSim**$^+$, describing the concept, its operation and the user-interface in general terms. Following the introduction is a detailed, task-oriented description of the implementation of **PolSARproSim**$^+$. This is naturally divided into a number of stages: receipt of parameters describing the forest and SAR imaging process, construction of the forest, and calculation and output of the SAR images. Forest construction and SAR imaging processes are described first and a list of required parameters is generated during the discussion. This list is summarized at the end of the section on implementation.

**PolSARproSim**$^+$is designed to be a command line tool, the inputs are passed via a text file. Following input of the parameters **PolSARproSim**$^+$performs the requested simulations, generating a stack of interferometric polarimetric SAR images, and writing them to file, in flat binary format.

To make **PolSARproSim**$^+$available on most machines the code has been written in ANSI standard C, and the source code may be compiled as required on different platforms using a simple Makefile. The source code has been modified significantly from the PolSARproSim version distributed with the PolSARpro v4.0 suite. A list of the major changes is included in the introduction section.

## 2. Introduction

The computer simulation of SAR images of forest is not a new technique and capabilities already exist in the USA, France, the Netherlands and Finland, which use accurate tree models. Details of the theory and techniques required to simulate forest SAR imagery have been published widely in the open literature, and the interested reader is referred to the reference section for further reading (particularly Cloude, Corr and Williams, 2004 [3], and Varenkamp, 2005 [14]). The availability of substantial computer power on a desktop or even notebook machine, coupled with efficiency of coding and algorithmic simplicity, has helped to bring this capability to the scientific community. The techniques described here follow a physics-based approach to the simulation of SAR imagery. The starting point is a detailed (but limited) physical description of the scene, including in addition to geometry such quantities as surface roughness and the frequency-dependent dielectric behavior of materials. Theoretical attenuation and scattering models for each element of the artificial 3D scene are combined with a model for coherent SAR imaging to construct images as coherent sums of focused scattering events. Images simulated in this way display properties consistent with those observed in real SAR imagery. In particular the simulated backscattering coefficients, speckle distributions, shadowing, polarimetric and interferometric coherence properties may all be simulated with a high level of realism. One should always remember that the simulated data is not real and contains many approximations. For example each scatterer is assumed to scatter the mean field, not all scattering processes are included, the attenuation suffered by the coherent field is modeled only approximately, and the images contain no system error artifacts other than thermal noise.

### 2.1. **PolSARproSim$^+$Concept.**

**PolSARproSim$^+$**is designed to overcome the limited availability of suitable data. The user is provided with an interface that permits a detailed description of a realistic forest model and a flexible description of SAR imaging parameters. Having designed the forest and set the imaging scenario the simulation is called from the command line, and in a short space of time (depending upon computer platform) the user has available simulated polarimetric, interferometric SAR data for the scene and sensor of their choice. In addition to the final polarimetric, interferometric SAR imagery, **PolSARproSim$^+$**generates additional files. For example a digital elevation map (DEM) file is generated, as is a list of stem locations and heights with crown dimensions, look vector files in the SCH coordinate system describing the location of the radar relative to each pixel etc.

### 2.2. **Capability Statement.**

**PolSARproSim$^+$**is called from the shell prompt as a standalone program to perform SAR imaging calculations of a model forest. Parameters controlling the operation of **PolSARproSim$^+$**are specified via a text file passed as a command line argument. **PolSARproSim$^+$**uses the parameter file values to construct a scene to be imaged, including a digital elevation map, and an array of trees, with specified surface and forest properties, such as ground roughness and the mean height of the forest stand. The description of the forest scene is reported. **PolSARproSim$^+$**then proceeds to

use the parameter file values to calculate a stack interferometric SAR images of the forest using those SAR sensor values and interferometric baseline requested by the user. These images are output in a binary format along with a record of ancillary information describing the imaging scenario, such as range to scene centre, centre frequency and baseline etc., into a text log file.

### 2.3. **Notable Features of PolSARproSim$^+$.**

Key features in the design of **PolSARproSim**$^+$are intended flexibility of scene and imaging parameters, maximum speed of execution, and fidelity of simulated imagery. **PolSARproSim**$^+$is written in ANSI C to help maintain portability across platforms. Memory requirements are minimized during execution by storing only a single realization of tree architecture at any time, and by treating small plant scattering elements using stochastic techniques that preserve interferometric coherence. Execution speed is optimized by using simplified forms for scattering by plant elements determined by size, and by modeling tree crowns as having uniform mean effective permittivity, thereby reducing the computational load associated with estimating the attenuation properties of the canopy, yet still preserving the inhomogeneity of the canopy in a realistic fashion. The code is written to take advantage of the multiprocessor architecture of modern computers by having multiple threads perform parallel computations during various stages of the simulation. Trees are represented and stored in parametric form, and realized using procedures coded according to chosen tree species, and dependent upon chosen tree heights through allometric equations. Tree realizations are subsequently divided into scattering elements for the SAR imaging calculation. A layer of understory vegetation is accommodated at a simple level. It is homogeneous and of constant depth above local ground level. The layer is comprised of small scattering elements, with uniform spatial and orientational distributions, yielding an azimuthally symmetric, homogeneous layer. A two-scale model is adopted for the underlying ground surface. The DEM generated using user-supplied parameters provides the large-scale surface. This is divided into facets, taking their orientation from the large-scale surface, and assigned small-scale roughness for the scattering calculation. Images are output in ground range and azimuth, and the backscattering alignment (BSA) convention is adopted.

### 2.4. **Flowchart for PolSARproSim$^+$Operation.**

**PolSARproSim**$^+$has an essentially linear design. Following input of the ASCII parameter file the forest scene is generated: firstly by constructing a large scale ground surface, and then populating the forest stand area with trees and short vegetation. The biophysical properties of the trees are then used to generate a 3D map of attenuation. The attenuation map is used as a look-up table in the final stage: the calculation and output of simulated fully polarimetric, interferometric SAR imagery. Figure 1 shows the **PolSARproSim**$^+$workflow for the major steps in the simulation process.

### 2.5. **Notable Changes to the Previous Version.**

**PolSARproSim**$^+$builds on the PolSARproSim version distributed with the PolSARpro v4.0 suite. In addition to all the capabilities that existed in the previous version, **PolSARproSim**$^+$includes the following features:

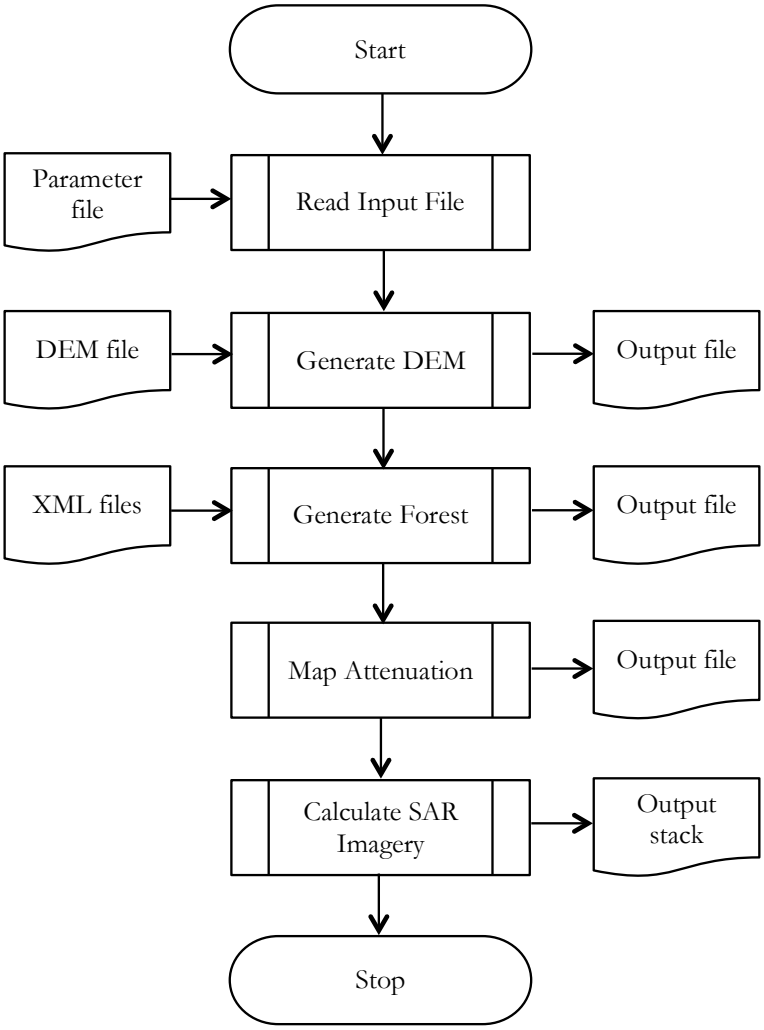


FIGURE 1. **PolSARproSim**$^{+}$workflow

- It is capable of simulating scattering from a forest stand with a mixture of species. The stand is set up using an XML file, where properties of each tree such as location, species, height, stem diameter and crown dimensions can be individually set.
- Unlike three hard-coded species in the previous version, any number of species can now be defined in significant detail using an allometry database file.
- The forest can be placed on an arbitrary topography which can be supplied using an external digital elevation model.
- Any number of tracks can be simulated by a single run of the software. The geometry of each track is specified as a baseline offset from the master track in the input text file.

- A more realistic and flexible system point spread function is implemented to allow various radar system configurations.
- The user can simulate the effects of temporal decorrelation by specifying motion and moisture changes for each track.
- The effects of thermal noise can be simulated. The user has control over the amount of thermal noise by specifying the noise equivalent sigma zero.
- The software architecture has been altered to take advantage of multiprocessor architectures. Each computationally intensive module has been re-written using POSIX threads, reducing run-time significantly.
- Further ancillary layers are now generated to aid data analysis, these include
  - The look vectors for each track in the SCH coordinate system.
  - The simulate terrain as a flat binary file.
  - The surface normal for each pixel.
  - Maximum heights seen in each pixel.
  - A map of pixels shaded from the radar due to topography as a flat binary file.
  - The three dimensional attenuation map.

## 3. General Considerations

### 3.1. **Terrain and Imaging Geometry.**

In the SAR simulation the imaging radar travels in the positive x-direction, with the antenna pointing port-side. The SAR imaging plane is the global $x_g, y_g$-plane ($z_g = 0$). The image area lies between $-L_x/2 \leq x_g \leq L_x/2$ and $-L_y/2 \leq y_g \leq L_y/2$ where $x_g$ is the azimuth coordinate, and $y_g$ is the ground range coordinate, which increases away from the platform trajectory. The forest stand occupies a circular area. In addition to the tree species (0-4), the mean height of the stand (meters), the stem density (trees/hectare) and the area of the forest stand (hectares) are supplied by the user. The scene area is calculated to comfortably accommodate the forest stand, taking into account layover and shadowing areas based on the user-supplied imaging geometry. Slant range and global incidence angle are defined with reference to the (horizontal) SAR image plane. The platform trajectory is assumed uniform and parallel to the SAR image plane. The large-scale ground surface, $g(x_g, y_g)$, is expressed as displacements about the mean terrain surface which is flat and tilted. The mean terrain surface has slope $s_x$ in the $x_g$ direction and $s_y$ in the $y_g$ direction. The ground surface height displacement about the mean terrain surface is periodic in $x_g$ and $y_g$. Imagery is output in the azimuth and ground range space, with far range appearing at the right of the images.

#### 3.1.1. *Imaging Geometry Calculations.*

In the example given in Section 3.1 the master track is specified with a slant range and incidence angle of

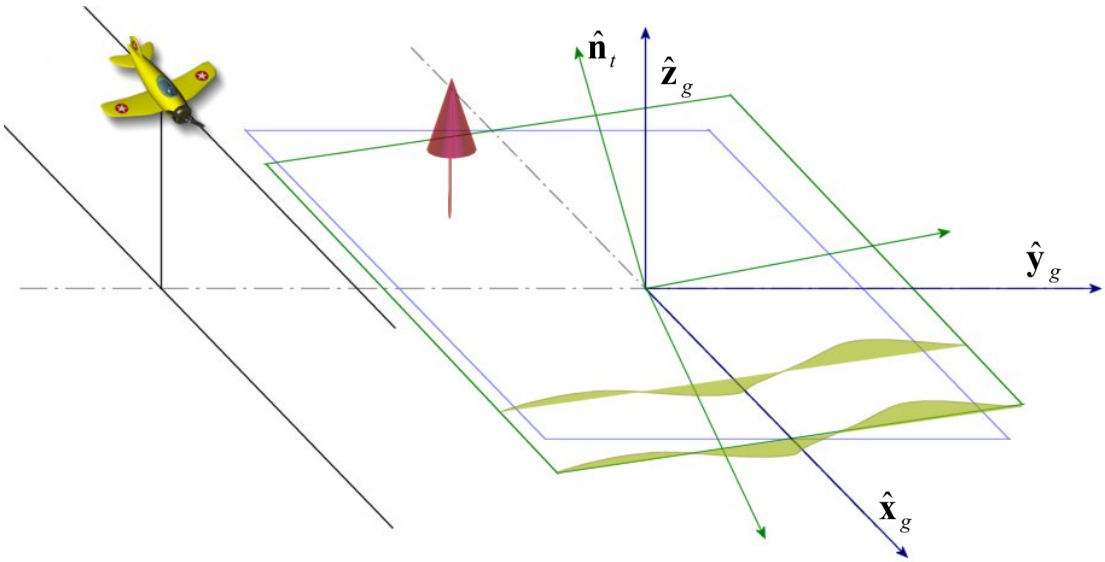


FIGURE 2. Terrain and imaging geometry

$$R_0 = 14433.756730\ m \tag{1}$$

$$\theta_0 = 40.0000°. \tag{2}$$

The platform height and ground range from the center of the scene ($x_g = 0, y_g = 0$) for the master track are therefore

$$H_0 = R_0 \cos\theta_0 = 11056.89914\ m \tag{3}$$

$$G_0 = R_0 \sin\theta_0 = 9277.839987\ m. \tag{4}$$

The first slave track is defined with a two dimensional baseline, see Figure 3, of

$$\vec{B}_1 = 100\hat{y} + 50\hat{z} \tag{5}$$

which yields platform height and ground range from the center of the scene for this track as

$$H_1 = H_0 + \vec{B}_1 \cdot \hat{y} = 11156.89914\ m \tag{6}$$

$$G_1 = G_0 + \vec{B}_1 \cdot \hat{z} = 9327.839987\ m. \tag{7}$$

The slant range and incidence angle to the scene center for the slave track are then

$$R_1 = \sqrt{H_1^2 + G_1^2} = 14542.52376\ m \tag{8}$$

$$\theta_1 = \tan^{-1}\left(\frac{G_1}{H_1}\right) = 39.8976^\circ \tag{9}$$

The slant range resolution has been specified as $\Delta y_s = 1.2\ m$, which translates to ground range resolutions of

$$\Delta y_0 = \frac{\Delta y_s}{\sin\theta_0} = 1.8660\ m \tag{10}$$

$$\Delta y_1 = \frac{\Delta y_s}{\sin\theta_1} = 1.8708\ m \tag{11}$$

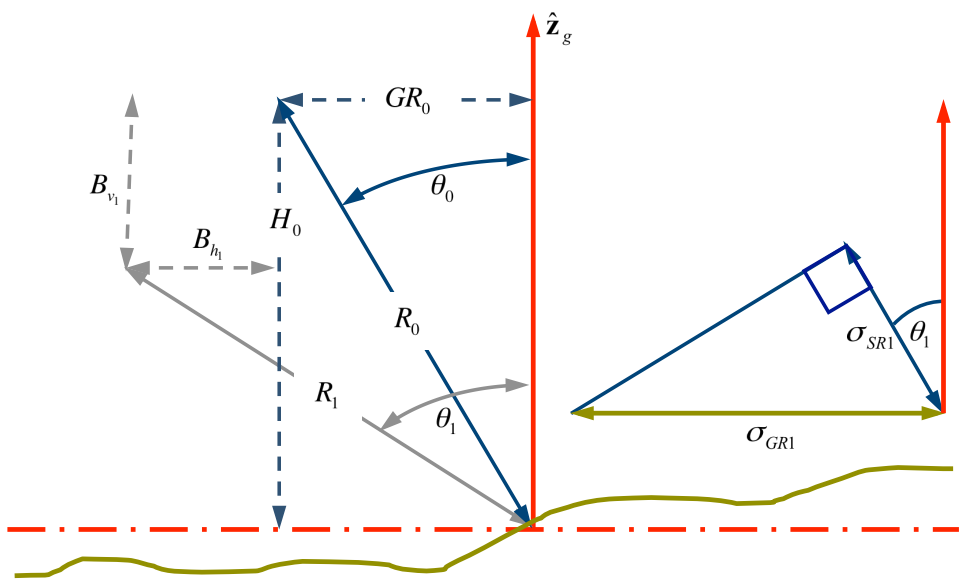


FIGURE 3. SAR imaging geometry

### 3.2. **SAR Image dimensions.**

The area imaged by **PolSARproSim**$^+$depends on the size of the forest, tree heights and the imaging geometry. Figure 4 shows the geometry of the imaged area.

The image dimension along the azimuth and ground range are given by

$$L_x = 2 \times (\text{Stand Radius} + \text{Gap} + \text{Max crown radius}) \tag{12}$$

$$L_y = 2 \times (\text{Stand Radius} + \text{Gap} + \text{Max crown radius}) + \text{Layover} + \text{Shadow} \tag{13}$$

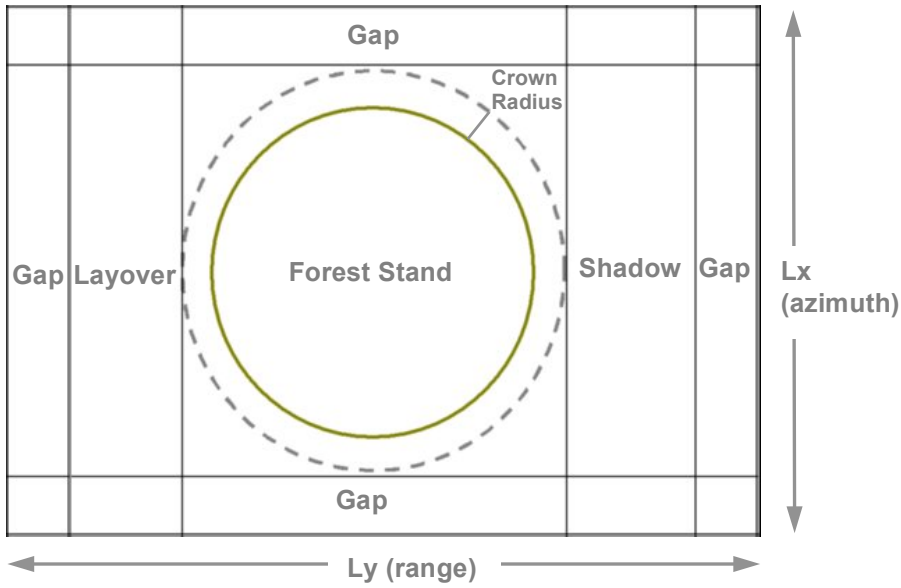


FIGURE 4. Scene geometry

where the regions marked in Figure 4 as Layover, Shadow and Gap are calculated as

$$\text{Layover} = (\text{Max tree Height})/\tan\theta_0 \tag{14}$$

$$\text{Shadow} = (\text{Max tree Height}) \times \tan\theta_0 \tag{15}$$

$$\text{Gap} = 5 \times \delta_y \times \delta_x. \tag{16}$$

Where $\delta_y$ and $\delta_y$ are the pixel dimensions in ground range and azimuth respectively, computed as

$$\delta_x = \text{ASF} \times \Delta x \tag{17}$$

$$\delta_y = \text{RSF} \times \Delta y \tag{18}$$

$$n_x = \lfloor L_x/\delta_x \rfloor + 1 \tag{19}$$

$$\delta_x = L_x/n_x \tag{20}$$

$$n_y = \lfloor L_y/\delta_y \rfloor + 1 \tag{21}$$

$$\delta_y = L_y/n_y \tag{22}$$

where ASF and RSF are the azimuth and ground range sampling factors respectively (specified in the input file), $\delta_x$ and $\delta_y$ are the pixel dimensions in meters and $n_x$ and $n_y$ are the number of pixels in azimuth and ground range.

## 4. Generation of the Simulated Terrain

This section describes the options for generating the simulated terrain that consists of underlying smooth topography and a rough surface. The user has the option of defining the smooth topography as either a tilted flat surface or by supplying an external DEM. Roughness is then added to the tilted surface or the externally read elevation model to complete the description of the simulated terrain. The tilted flat surface is described using two parameters in the input file, namely '`ground_slope_azimuth`' and '`ground_slope_range`', the slopes in azimuth and range respectively. Alternatively, an external DEM can be supplied as a binary file.

### 4.1. **Reading in an External DEM.**

The ground elevation, as described in Section 3.1, is assumed to be a real offset from a plane at zero height. If chosen, the tilted surface intersects that plane at the center of the image. An external DEM, however, allows the user flexibility to define arbitrary heights anywhere in the scene relative to the zero elevation plane. It is assumed that the center of the DEM is aligned with the center of the simulated image, and that they are co-registered (i.e. there is no misalignment). Parameters of the DEM file (assumed to be a raster double precision binary file) such as the number of pixels in range and azimuth and the respective resolutions are provided in the input parameter file (see Section 7.2 for details). Once read, **PolSARproSim**$^{+}$trims then resamples the DEM to the simulation coordinates using a simple bilinear interpolation. Since the DEM is resampled, its resolution can be changed by simply editing the respective parameters in the input file. In order to account for pixels occluded from the radar because of large slopes, a map of the shadow heights is then generated using geometry from the master track. This 2-D map contains the height up to which an object in that particular location would be hidden from the radar. Figure 5 illustrates the shadow heights ($h_{sh}$)

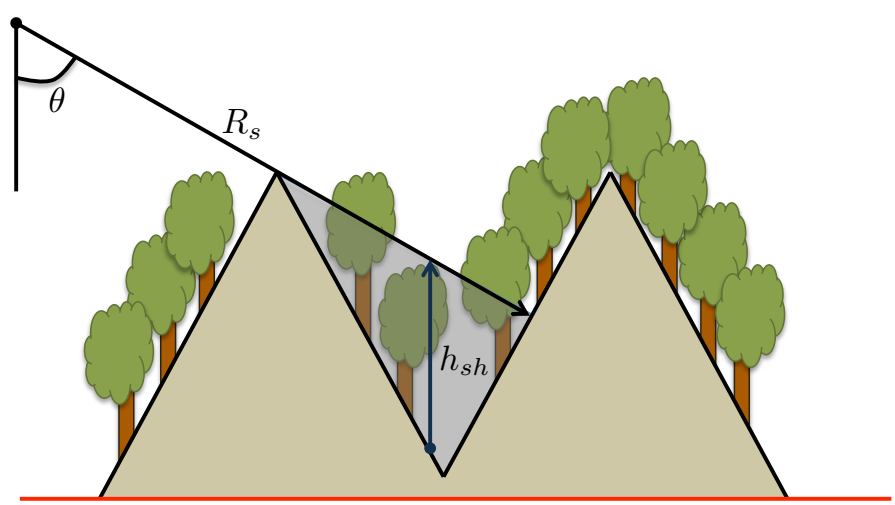


Figure 5. Illustration of a shadow map heights

computed for each pixel using the master track geometry (i.e the slant range, $R_s$, and incidence angle, $\theta$). The shadow heights are then used to estimate whether a tree (or parts of it) that is placed on the topography will be occluded form the radar using the attenuation map (described in Section 6.2).

4.2. **Generating the Rough Surface.**
Once the DEM (either a sloped plane or the external DEM) has been created, a rough undulating surface that is then draped over the existing DEM. This section describes the techniques used for the generation of the large-scale roughness model. Arguments are given for the 2D case first, and then extended to 3D. The methods are inspired by those described in [12]. The desired surface realization is described using a truncated Fourier series, given by

$$h(x) = \frac{1}{L}\sum_{n=-N}^{N} h_n \exp\left(i2\pi nx/L\right) \tag{23}$$

where $L$ is the length of the surface to be generated, and the surface is periodic with length $L$ such that $h(nx+L) = h(x)$. Now writing $x' = 2\pi x/L$ we have

$$h(x') = \frac{1}{L}\sum_{n=-N}^{N} h_n \exp\left(inx'\right) \tag{24}$$

$$h(nx'+L) = h(x') \tag{25}$$

$$h_m = \frac{L}{2\pi}\int_0^{2\pi} h(x')\exp\left(-imx'\right)dx' \equiv H\left(k_m = 2m\pi L\right). \tag{26}$$

The underlying surface correlation function is

$$c_h(s) = \int h(x)(x+s)\,dx \tag{27}$$

Suppose we have a random function $f(x)$ with a very narrow correlation function. A realization of such a function may be generated with a suitable random number generator. The Fourier transform (FT) of our desired function is

$$H(k) = W(k)F(k) \tag{28}$$

where $F(k) = \mathrm{FT}\left[f(x)\right]$. The desired correlation can then be written

$$c_h(s) = \int h(x)(x+s)\,dx \tag{29}$$

$$= \mathrm{FT}^{-1}\left[\left|H(k)\right|^2\right] \tag{30}$$

$$= \mathrm{FT}^{-1}\left[\left|W(k)\right|^2\left|F(k)\right|^2\right] \tag{31}$$

In like manner the correlation of our random function is expressed as

$$c_f(s) = \mathrm{FT}^{-1}\left[|H(k)|^2\right] \tag{32}$$

This correlation is very narrow, and in the limit of completely random noise we expect $c_f(s) \to \delta(s)$, when we have, after suitable normalization

$$C_h(k) = \mathrm{FT}\left[c_h(s)\right] = |W(k)|^2 \tag{33}$$

from which it follows that

$$H(k) = \left[C_h(k)\right]^{1/2} F(k) \tag{34}$$

or in the discrete model

$$H(k) = \left[C_h(k)\right]^{1/2} F(k_m) \tag{35}$$

$$H(k_m = 2m\pi L) = \frac{L}{2\pi}\int_0^{2\pi} h(x')\exp(-imx')\,dx' = h_m \tag{36}$$

Now the slopes on the mean flat terrain surface are $s_x$ and $s_y$ such that for any point in the SAR image plane $r = (x, y, 0)$ , the height of the mean flat surface is given as $xs_x + ys_y$. The mean terrain surface has surface normal vector

$$\hat{n}_t = \frac{(\hat{z}_g - s_x\hat{x}_g - s_y\hat{y}_g)}{\sqrt{1 + s_x^2 + s_y^2}} \tag{37}$$

which makes an angle $\theta_t$ with the global z-direction $\hat{z}_g$ such that

$$\cos\theta_t = \hat{n}_t \cdot \hat{z}_g = \frac{1}{\sqrt{1 + s_x^2 + s_y^2}}. \tag{38}$$

The displacements $h(x, y)$ calculated using the technique described in the following text are in the direction $\hat{n}_t$ away from the mean surface at $t = (x, y, xs_x, ys_y)$. To convert to the global frame make the simple approximation that the displacement $h$ in the direction $\hat{n}_t$ corresponds to a displacement in the global z-direction $\hat{z}_g$ of $h\sqrt{1 + s_x^2 + s_y^2}$. Thus the final height of the ground surface at azimuth $x$ and ground range $y$ is:

$$g(x, y) = xs_x + ys_y + h(x, y)\sqrt{1 + s_x^2 + s_y^2} \tag{39}$$

4.2.1. *Roughness Generation Algorithm.*
The process of constructing the realization of the surface proceeds as follows (now in 3D). The first step is to create a random function $f(x, y)$ in the frame of the local sloping terrain, then determine the coefficients

$$F\left(k_m, k_n\right) = \int_0^{2\pi} \int_0^{2\pi} f\left(x', y'\right) \exp\left(-imx'\right) \exp\left(-iny'\right) dx' dy' = f_{mn} \tag{40}$$

where the lengths are increased by a factor $a = \sqrt{1 + s_x^2 + s_y^2}$. The overall scaling can wait until the end. The integrals may be approximated as discrete sums for convenience

$$F\left(k_{xm}, k_{yn}\right) = \sum_{i=0}^{N} \sum_{j=0}^{M} f\left(x_i', y_j'\right) \exp\left(-imx_i'\right) \exp\left(-iny_j'\right) dx' dy' = f_{mn} \tag{41}$$

where $x_i' = i2\pi\Delta x/L_x$ and $y_i' = i2\pi\Delta y/L_y$, and all lengths are in the local terrain frame. Note here that $-N \leq m, n \leq N$ and $k_{xm} = 2m\pi/Lx$, while $k_{yn} = 2n\pi/Ly$. Then zero the mean height, $h\left(x, y\right) \rightarrow h\left(x, y\right) - \bar{h}$, and normalize the standard deviation to the desired value, $h\left(x, y\right) = \alpha h\left(x, y\right)$. Finally the heights $(x, y)$ may be interpreted as displacements about a tilted ground plane, and converted to global SAR frame displacements using

$$g\left(x_m, y_n\right) = x_m s_x + y_n s_y + h\left(x_m, y_n\right) \sqrt{1 + s_x^2 + s_y^2}. \tag{42}$$

Note that with the choice

$$c_h\left(s\right) = \exp\left(-s^2/l^2\right) \tag{43}$$

and using

$$C_h\left(k\right) = \frac{1}{\sqrt{2\pi}} \int_{\infty}^{\infty} c_h\left(s\right) \exp\left(iks\right) ds \tag{44}$$

we find

$$\sqrt{C_h\left(k\right)} = \left(\frac{l}{\sqrt{2}}\right)^{1/2} \exp\left(-k^2 l^2/8\right) \tag{45}$$

Note that the algorithm yields a periodic set of displacements. To calculate the ground height outside of the SAR image area

$$g\left(x + mL_x, y + nL_y\right) = g\left(x, y\right) + mL_x s_x + nL_y s_y. \tag{46}$$

4.2.2. *Algorithm Inputs.*
The following parameters are required to generate the large-scale DEM. The image area dimensions in meters, $L_x$ and $L_y$, generated from the forest area using knowledge of the imaging geometry etc. Image dimensions in pixels, $n_x$ and $n_y$, depending upon encoded pixel sampling rate, resulting from specified resolutions and image dimensions in meters. The number of terms in Fourier expansion, $N$, specified by the programmer. The terrain slopes, $s_x$ and $s_y$, under control of the user or the external DEM. Large scale surface correlation length in meters, $l$, recovered from mean tree separation. Larg-scale

surface height standard deviation in meters, $\sigma_l$ , recovered using a roughness model. The user chooses between surface roughness on the basis of, "smoothÓ - 0, to "roughÓ- 10. Roughness depends upon wavelength and incidence angle. Thus the userÕs choice of surface roughness value (0-10) translates, into a numerical value for surface height standard deviation in meters.

4.2.3. *Choice of Large-Scale Surface Roughness.*
Large-scale roughness is linked to power loss in the specularly reflected wave via the Rayleigh parameter [10] which scales the power of the specularly reflected wave according to

$$|R|^2 = \exp\left(-4k^2\sigma_t^2\cos^2\theta\right) \tag{47}$$

where $\sigma_t = \sigma_s + \sigma_l$, is the total (large-scale + small-scale) roughness. The user enters a roughness number $0 \leq r \leq 10$ from which the power reduction on the reflected wave is determined according to

$$10\log_{10}\left(|R|^2\right) = \min R_{dB} - \left(\frac{r}{10}\right)(\min R_{dB} - \max R_{dB}) \tag{48}$$

where $\min R_{dB}$ and $\max R_{dB}$ are programmer assigned constants. The largeÐscale roughness is then assigned according to

$$\sigma_l = \frac{\sqrt{\ln\left(|R|^2\right)}}{2k\cos\theta} - \sigma_s \tag{49}$$

The large-scale correlation length is always chosen as 3 times the mean tree crown radius. The choice of small-scale roughness and correlation length is discussed in the section on SAR image calculation.

## 5. Generation of the Simulated Forest

This section describes the options for generating the simulated forest stand. There are two aspects to the description of the forest, a) tree by tree primitives such as location, species, height, stem diameter, canopy dimensions and b) the realization of each tree based on species specific allometric parameters. The tree by tree primitives can either be supplied through an input XML database file (see Section 7.3 for details) or **PolSARproSim**$^+$can be tasked to generate random tree locations (and other tree primitives based on species allometry) for a forest stand of a single species and a mean height. In each case, the tree locations are a list of azimuth and ground range coordinates for the base of the stem of each tree relative to scene center. For each tree, then, the stem, primary, secondary, tertiary branches and leaves are realized based on species specific allometric parameters supplied through an input XML database file (see Section 7.4 for details). The scattering from stems and branches is implemented by dividing each element into small segments or cylinders, each with a distinct orientation and position that preserves forest structure. While the segments or cylinders

are created only when they are imaged, the trees are stored in memory as data structures with the tree primitives (location, species, heights etc) and parameters for all the realized stems branches in linked lists.

### 5.1. **Internally Generated Forest Parameters.**

This section describes the algorithms used by **PolSARproSim**$^{+}$when it is tasked to randomly place trees of a species inside a circular stand of a specified area instead of explicitly providing forest parameters through the input XML file. Locations for each coordinate pair and an associated tree height are first drawn from a normal distribution. The user specifies directly the mean tree height in meters. The tree height standard deviation is set to 5% of the mean value. The tree height is a fundamental quantity. All allometric equations are functions of tree species (specified by the user through the allometry XML file) and tree height. Thus the tree global crown radius (see Section 5.2.1 for definition of the Tree "object"), is calculated from tree height using a species-specific allometric equation. This crown radius is used in the generation of tree locations. The process of tree location map generation begins by determining the number of trees in the scene, and then initializing them in a regular pattern, and realizing their heights and global crown radii. The trees are subsequently "shuffled" around in random, collision-avoiding walks using Monte Carlo techniques, to reach a more realistic distribution of tree positions.

The area, tree species, and mean height of the forest stand are all provided by the user. Crown dimensions are derived from the last two quantities using allometric equations. Given the area and the mean crown radius the maximum number of trees that might be expected within that area if they are close-packed in 2-dimensions (with discs of mean crown radius occupying a fraction 0.907 of the total area) is calculated. Trees are permitted to have crowns that overlap by as much as 11%. The user specifies stand density in stems/hectare. If the requested stand density is greater than the calculated maximum by a certain amount the actual stand density is the smaller of the two, otherwise the actual stand density is the requested density. From the actual stem density in stems/hectare the stem line density in stems/100m is determined as the square root of the stem density. Initially trees are to be located on grid coordinates, and the grid dimensions are now recovered using the area dimensions and the stem line density. Note that since the final number of trees is the product of two integers, so the requested stem density is not matched precisely. Having determined the grid dimensions the trees are located on grid points with alternate lines shifted by half a grid spacing to offset the initial tree positions. Tree locations are then "shuffled" using an annealing Monte Carlo algorithm [1]. The shuffling is controlled using the crown radii and a cost function, which rises steeply as the trees approach to a separation within their combined radii, and has a small attractive effect for greater tree separations. The technique yields realistic, non-uniform distributions of tree locations, which appear quite natural. The shuffling treats the image area as periodic, so that trees shuffling out of the area to the right, shuffle into the area from the left, and similarly from top to bottom, and vice versa. Exploiting periodicity permits the expansion of the simulated area, whilst preserving a natural tree separation distribution, and avoiding trees that are too close together. Finally, from the full list of tree positions, a circular

area is excised that corresponds to the forest stand area requested by the user. This circular area is offset slightly in ground-range in order to accommodate the extent of layover. The tree locations, heights and crown radii are output to a text file. The scene geometry accommodates layover and shadow regions as calculated previously.

In summary the following parameters are required to generate the tree location map: the forest stand area, the mean tree height in meters, and the requested stand density. In addition the allometric equation relating tree mean crown radius to tree height and species is also used.

### 5.2. **Tree Species and Architecture.**

To complete the description of the forest stand we require a description of the tree architectures. Each tree, when realized, appears unique: with its own arrangement of stem, primary and secondary branches, and a crown volume filled with tertiary branch elements and leaves. The description of the tree then resides in the procedures and algorithms used to generate these realizations. A broad description of these algorithms, and the structures upon which they operate, is given in this section. Actual parameter values controlling their operation are specified in the allometry XML file which is described in detail in Section 7.4. An example file contains parameter for a couple of species include values that have been refined during the development process to ensure that predicted SAR backscattering coefficients are within observational limits.

#### 5.2.1. *Trees.*

Although programming in C it is useful to think, in object-oriented terms, of trees represented in the software as objects, and in practice trees are represented as structures. The definition of a tree record includes the species, a vector location, stem diameter, tree height, and canopy dimensions in meters. These quantities are either specified in the forest XML file or assigned and used in the tree shuffling procedure. In addition the record contains lists of stems, primary and secondary branch objects, whose definitions are considered shortly. The definition of a tree record also contains a list of crown volumes occupied by tertiary branches and leaves (or needles).

#### 5.2.2. *Crowns.*

All crown objects relate to volumes of type cylinder, cone or truncated ellipsoid. Crowns are populated by leaves and twigs (tertiary branches) and these require dimensions, permittivities, orientations and volume fractions, that are all specified in the allometry XML file by species. At this time only two crowns are accommodated per tree, ostensibly a crown of living branches and one with dead branches called the dry crown. The living crown is assumed to start at the top of the tree defined by its shape, depth and radius. The dry crown is assumed to exist below the living crown, inherits its radius form the living crown while its depth must be specified. Crowns are truncated by bounding planes that are always taken parallel to the mean sloping terrain. Since they are known to pass through specific points they are not stored explicitly as part of Crown structures. Instead Crown structures store the slopes of the bounding planes, and Plane objects are constructed from crown field data as required. It is necessary to understand the intersection of rays with planar and crown surfaces

in order to implement the estimation of attenuation when performing the SAR calculation. So in addition to a planar surface a Ray object is also defined. The details of Ray-Crown and Ray-Plane intersections are given in Appendix A.

5.2.3. *Branches.*
Branches are represented in the computer as curved, tapering cylinders. These are converted into a series of short, straight cylindrical sections for the purposes of the scattering calculation. The basic branch shape is defined in terms of the location of the centre of the branch

$$\vec{b}\,(t) = b_0 + l\left[t\hat{z}_0 + \frac{d_p}{2\sqrt{1-d_p^2}}t^2\hat{p} + c\,(t)\right] \tag{50}$$

where $l$ is the "straight" branch length, and $t$ is a parametric value, which is zero at the start of the branch and unity at the tip of the branch. The unit vector $\hat{z}_0$ defines the initial direction of the branch. In the absence of shape variation, $\vec{c}\,(t)$, and curvature the branch is straight and $\vec{b}\,(t) = \vec{b}_0 + lt\hat{z}_0$. The branch may bend towards a specific direction, $\hat{p}$, to mimic the natural tendency of branches to grow towards the light or to sag under the effects of gravity. For example see [15, 9] (Weber and Penn, 1995, Lintermann and Deussen, 1999). A quadratic form for the bending function is adopted such that in the absence of shape variation the direction of the branch is

$$\hat{z}\,(t) = \frac{\left(1-d_p^2\right)^{1/2}\hat{z}_0 + td_p\hat{p}}{\sqrt{1+\left(t^2-1\right)d_p^2}} \tag{51}$$

where $0 \leq d_p \leq 1$ is a "bending factor" (species and branching order dependent) controlling the amount of bending towards the $\hat{p}$ direction. The shape variation is chosen to mimic the irregular shapes of real branches and is zero for values of the parameter $t = 0$ and $t = 1$. The branch crookedness, $\vec{c}\,(t)$ is modeled as

$$\vec{c}\,(t) = \hat{x}_s a_x + \hat{y}_s a_y \tag{52}$$

where

$$\hat{x}_s = \hat{z} \times \hat{z}_s \tag{53}$$

$$\hat{y}_s = \hat{x}_s \times \hat{z}_s \tag{54}$$

$$\hat{z}_s = \frac{\hat{z}_0\sqrt{1-d_p^2} + \hat{p}d_p t}{\left|\hat{z}_0\sqrt{1-d_p^2} + \hat{p}d_p t\right|} \tag{55}$$

$$\begin{aligned} a_x = &-\frac{\gamma\lambda_{cx}}{2\pi}\left(\cos\left(\frac{2\pi t}{\lambda_{cx}} + \phi_x\right) - \cos\phi_x\right) \\ &- \frac{\alpha_G\gamma\lambda_{cx}}{2\pi}\left(\cos\left(\frac{2\pi\alpha_G t}{\lambda_{cx}} + \phi_x\right) - \cos\phi_{cx}\right) \end{aligned} \tag{56}$$

$$\begin{aligned} a_y = &-\frac{\gamma\lambda_{cy}}{2\pi}\left(\cos\left(\frac{2\pi t}{\lambda_{cy}} + \phi_y\right) - \cos\phi_y\right) \\ &- \frac{\alpha_G\gamma\lambda_{cy}}{2\pi}\left(\cos\left(\frac{2\pi\alpha_G t}{\lambda_{cy}} + \phi_y\right) - \cos\phi_{cy}\right) \end{aligned} \tag{57}$$

where $\alpha_G$ is the golden ratio, 1.61803399, while $\gamma, \lambda_{cx}, \lambda_{cy}$ modulate the amount of crookedness and the periodicity of the crookedness of a branch. They are under the control of the user, while values for parameters $\phi_x, \phi_y, \phi_{cx}, \phi_{cy}$ are drawn from species-specific distributions. For example the start radius depends upon the radius of the parent branch at the branching point etc. The branch radius decays linearly from the start value to the end value with the parameter, $t$. Branches are subdivided into segments (Cylinders) in the code for drawing and scattering calculations, and the branch equation is used to determine the Cylinder properties for each branch segment. The electrical properties of those cylinders (in essence the effective permittivity) are estimated as a function of the gravimetric moisture content ($M_g$) using [13]

$$\epsilon_v = \epsilon_r + \epsilon_f v_{fw} + \epsilon_b v_b \tag{58}$$

where $\epsilon_v$ is the effective permittivity of the bulk vegetation, $\epsilon_r$ the residual dielectric constant, $v_{fw}\epsilon_f$ a fraction of free water and its permittivity while $v_b\epsilon_b$ the bulk vegetation bound water component and its permittivity. The residual dielectric constant is modeled as

$$\epsilon_r = 1.7 - 0.74M_g + 6.16M_g^2 \tag{59}$$

while the vegetation fractions are given by

$$v_b = 4.64\frac{M_g^2}{1 + 7.36M_g^2} \tag{60}$$

$$v_{fw} = M_g\left(0.55M_g - 0.076\right) \tag{61}$$

with their respective permittivities given by

$$\epsilon_f = \left[\epsilon_{f\infty} + \frac{\epsilon_{fs} - \epsilon_{f\infty}}{1 + (f/f_{f_0})^2}\right] - j\left[\frac{f/f_{f_0}\left(\epsilon_{fs} - \epsilon_{f\infty}\right)}{1 + (f/f_{f_0})^2} + \frac{g\sigma}{2\pi\epsilon_0 f}\right] \tag{62}$$

$$\epsilon_b = \epsilon_{b\infty} + \frac{\epsilon_{bs} - \epsilon_{f\infty}}{1 + \sqrt{jf/f_{b_0}}} \tag{63}$$

where $\epsilon_{f\infty} = 4.9$ and $\epsilon_{fs} = 80.88$ are the high-frequency and static limits of the saline water dielectric constant. The radar frequency is given by $f$ (in Hertz) while $f_{f_0} = 17.157\ GHz$ is the relaxation frequency at $22°C$. The ionic conductivity parameter $\sigma$ in Siemens/meter is assumed to be 1.27 while the free space permittivity is $\epsilon_0 = 8.854 \times 10^{-12}\ F$. The high-frequency and static limits of the bound water are $\epsilon_{bs} = 57.9$ and $\epsilon_{b\infty} = 2.9$ while the corresponding relaxation frequency is $f_{b_0} = 0.18\ GHz$.

### 5.3. **Realization of Trees.**

Before a tree is imaged in **PolSARproSim**$^+$it is realized. Realization refers to the populating the linked lists containing geometric parameters describing crowns, stems and branches (such as in (50)) as well as the electrical parameters for each scattering object. The first step in the process of realizing a tree is to populate the list of crowns. The Branch lists, describing the stem, primary and secondary and tertiary branches are then created and populated based on the crown dimensions and allometric parameters. Finally, leafs are placed randomly throughout the living crown and their dimensions and locations are saved in Foliage lists.

#### 5.3.1. *Generating Stems and Branches.*

The definition of a branch is based around an equation describing the curved branch centre. This equation has the form (repeated here for convenience)

$$\vec{b}\,(t) = b_0 + l\left[t\hat{z}_0 + \frac{d_p}{2\sqrt{1 - d_p^2}}t^2\hat{p} + c\,(t)\right] \tag{64}$$

where the non-curved "direction" of the branch is

$$\hat{z}\,(t) = \frac{\left(1 - d_p^2\right)^{1/2}\hat{z}_0 + td_p\hat{p}}{\sqrt{1 + (t^2 - 1)\,d_p^2}} \tag{65}$$

The curvature component is a correction to the central branch position, and parameter values are chosen appropriately (species and branching order dependencies are involved). To find the remaining parameter, $l$, specify the branch termination on the crown volume surface, i.e.

$$\vec{b}\,(t=1) = \vec{b}_0 + l\left[\hat{z}_0 + \frac{d_p}{2\sqrt{1-d_p^2}}\hat{p}\right] = \vec{b}_0 + l\,[\hat{z}_0 + f_p\hat{p}] \tag{66}$$

This point is found by starting at the point $\vec{b}_0$, and searching for the intersection of a ray with the crown volume in the direction $\hat{a} = (\hat{z}_0 + f_p\hat{p}) \,/\, |\hat{z}_0 + f_p\hat{p}|$ to find a point, $\vec{b}_l$, from which the "straight length", $l$, may be recovered as

$$l = \frac{\left(\vec{b}_1 - \vec{b}_0\right)\hat{z}_0}{1 + f_p\,(\hat{p}\cdot\hat{z}_0)} \tag{67}$$

where

$$f_p = \frac{d_p}{2\sqrt{1-d_p^2}} \tag{68}$$

We specify the base, initial direction and bending vectors at the outset, then the branch length is found by seeking the intersection of the ray with the crown surface. The same process is adopted for stems, primaries and secondary branches. Thus, given the appropriate rules, and techniques for finding ray intersections with crown volume surfaces, we have all that is necessary to construct the Trees. During the creation of secondary branches temporary sub-crowns can be created for each primary branch. The algorithms for finding ray intersections with different geometric objects are given in the Appendix A.

## 6. Interferometric SAR Image Calculation

It might appear to be most efficient to calculate both images at the same time, using a "dual-antenna" approach rather than a "repeat-pass" approach. This would have the advantage of only requiring realization of the forest a single time. However there are a number of overheads associated with this idea. The first and most obvious is the memory requirement. Not only is this doubled for the images themselves but other structures, such as the attenuation grid, to be discussed in the following sections, must be reproduced. Nor is there a great saving in computational load. Whilst the "repeat-pass" method doubles the realization cost, and requires care and attention to exact reproduction of the scene, this computational load is not a critical factor. From experience the heaviest computational load arises in the calculation of effective permittivity by voxel, scattering amplitude calculations using sophisticated models for cylinder scattering, and determination of focus points, particularly for the ground-volume interactions. All of these overheads are present in both the "dual-antenna" and "repeat-pass" approaches. Thus the approach to be adopted is "repeat-pass", which in some respects presents a less complex coding challenge, with a greater concomitant potential for success. This involves calling a single routine twice, each time with a slightly different imaging geometry. The SAR image calculation phase may be

broken down into stages. The first of these is the calculation of canopy tree crown and understory permittivities. Crowns are taken to have the same mean permittivity. Thus tree crown effective permittivities are calculated in the Foldy-Lax approximation [11] by averaging the forward scattering amplitudes of their constituent elements. The same approach is adopted for the short vegetation layer, which is also taken to be homogeneous and isotropic, and to have constant depth above local ground height. The next stage is to construct attenuation grids, which supply look-up tables for attenuation by spatial location. The attenuation grids are used when calculating scattering to provide approximate estimates of attenuation by location in a very efficient way that enhances the simulation performance, yet still maintain the effects of the inhomogeneity of the tree crown distribution. Having determined the attenuation grids the next stage is the calculation of SAR images. This starts with the direct-ground (DG) calculation, which employs the techniques described in [16](M L Williams, 2006) and which are reproduced in the following sections. Essentially the ground surface is divided into numerous facets, and these are assigned complex scattering amplitudes, and focused in the image one after the other, each being scaled by the attenuation estimate for its location in the scene. The next calculations, direct-volume (DV) and ground-surface-volume (SV) for the short vegetation layer are conducted together to reduce computational load. The technique uses the hybrid stochastic-deterministic approach described in [16]. A subset of plant elements is realized, and their scattering amplitudes scaled to yield the backscattering coefficient anticipated by the full sample. The number of realizations per resolution cell is sufficient to provide fully-developed speckle, and the actual realizations are preserved for the second interferometric SAR image. The final calculation is conducted for the forest of trees. Each tree location is addressed in turn. A unique distribution of stem, primary and secondary branches is created and subdivided into short cylindrical elements. Both direct-forest (DF) and ground-surface-forest (SF) terms are calculated for these elements. Finally the hybrid technique adopted for short vegetation is adopted for scattering from the smaller tree crown elements of leaves, twigs and pine needles. After writing out the first set of SAR imagery to an appropriately named file, the whole process is repeated for the repeat-pass scenario, taking care to preserve random number sequences where appropriate and ensure consistency of scene realization. The full process is described in greater detail in the following sections.

### 6.1. **Representation of the SAR Image.**

The forward SAR model employs a representation of the SAR image as a coherent (phase-preserving) transform of a radar reflectivity function. This model is derived in [16], and the derivation is repeated here for convenience. The SAR image is taken to be formed using transmission and reception of regular pulses of the form

$$\vec{p}\,(\tau) = \int_{\infty}^{\infty} \hat{p}\,(\omega)\, e^{-i\omega\tau} d\omega\ \hat{e} = p\,(\tau)\,\hat{e} \tag{69}$$

The received signal with the platform at "position" $x$ is

$$\vec{\sigma}(\tau, x) = \int_{\infty}^{\infty} \underline{\vec{B}}(\omega, x) \cdot \hat{e}\left(\hat{p}(\omega) e^{-i\omega\tau}\right) d\omega \tag{70}$$

i.e. the total return from the illuminated scene, where

$$\underline{\vec{B}}(\omega, x) = \int_{\infty}^{\infty} \underline{\vec{F}}(\omega, x, s) e^{i\omega 2r(x,s)/c} ds \tag{71}$$

and $\underline{\vec{F}}(\omega, x, s)$ is the reflectivity density at the point $s$, with range $r(x, s)$. The correlation of the received signal with the return anticipated from a delayed pulse is

$$\underline{\vec{I}}(t, y) = \int \vec{\sigma}(\tau, x, s) p^*(\tau - \tau'(x, t, y)) d\tau dx \tag{72}$$

$$= \int \vec{\sigma}(\tau, x, s) p^*\left(\tau - \frac{2\left|\vec{r}_p(x) - \vec{r}(t, y)\right|}{c}\right) d\tau dx \tag{73}$$

Here $\vec{r}_p(x)$ is the platform location at $x$, and $\vec{r}(t, y)$ is the location of the scattering center at the point of interest in the image. Using the definition of the pulse in terms of its Fourier spectrum

$$p^*(\tau - \tau') = \int_{\infty}^{\infty} \hat{p}^*(\omega') e^{i\omega'(\tau - \tau')} d\omega' \tag{74}$$

in the expression for the correlated image leads to

$$\underline{\vec{I}}(t, y) = \int \int_{\infty}^{\infty} \underline{\vec{B}}(\omega, x) \left|\hat{p}(\omega)\right|^2 e^{-i\omega\tau'(x,t,y)} d\omega dx \tag{75}$$

Substituting for $\underline{\vec{B}}$ the SAR image function is

$$\underline{\vec{I}}(t, y) = \int \int \int_{\infty}^{\infty} \underline{\vec{F}}(\omega, x, s) \left|\hat{p}(\omega)\right|^2 e^{i\omega(2r(x,s)/c - \tau'(x,t,y))} d\omega dx ds \tag{76}$$

The approximation of constant reflectivity across the SAR bandwidth and aperture is adopted so that

$$\underline{\vec{I}}(t, y) = \int \underline{\vec{F}}(s) \hat{Q}(t - r(s), y - x) ds \tag{77}$$

and

$$\hat{Q} = \int \int_{\infty}^{\infty} \left|\hat{p}(\omega)\right|^2 e^{i\omega(2r(x,s)/c - \tau'(x,t,y))} d\omega dx \tag{78}$$

is the system impulse response or point spread function. This quantity is approximated in the simulation using a Gaussian function product in azimuth and ground range, with appropriate range phase. This result is valid for direct scattering from a

surface or volume scattering, and higher orders of scattering involving surface-surface, or volume-surface, or surface-volume-surface etc. The point, $s$, need not be a location either in or on a scattering object, but rather in general it is an *effective* scattering center associated with the reflectivity determined from interacting scene elements that are spatially separated. A discrete approximation is employed

$$\underline{\vec{I}}\left(r', x'\right) = \sum_{n} \underline{\vec{F}}_n \hat{Q}\left(r' - r\left(\vec{s}_n\right), x' - x\left(\vec{s}_n\right)\right) \tag{79}$$

where the polarimetric scattering amplitude, $\underline{\vec{F}}_n$ , associated with the $n^{th}$ scattering event, arising from one or more elements of the scene, and incorporating attenuation, has an effective scattering centre, $\vec{s}_n$. SAR image prediction using this model proceeds as follows: describe the scene geometrically and physically, as described in the preceding sections, then divide the scene into small (relative to system resolution) elements. Interrogate the scene and determine scattering events (divided by class), calculating their polarimetric scattering amplitudes, and associated *effective* scattering centers. Sum their contributions coherently into the image accumulator, $\underline{\vec{I}}\left(r', x'\right)$. Whilst simple in concept the procedure entails a heavy computational load mitigated only by the use of approximations that preserve important features of the SAR imagery such as clutter distribution, shadowing and interferometric coherence.

### 6.2. **Modelling Attenuation.**

This requires estimation of the complex permittivity of the random media comprising the tree crowns and the short vegetation. In previous work of this kind the technique was to divide the canopy into voxels and to identify the material occupying those voxels. The scattering properties of occupying vegetation elements were used to estimate the mean, or effective permittivity in each voxel. Then during the SAR calculation, for each scattering event a list was built indicating through which voxels the broadside ray passed for that event, and the attenuation due to each voxel was calculated and accumulated into the scattering amplitude. This process is somewhat inefficient and therefore unsuitable to the present task. The intention is to adopt a new, more efficient method. The first stage is to estimate an effective permittivity for both the short vegetation layer, and the tree crowns. Each crown is given the same mean permittivity. The permittivity estimate is determined by integrating the forward scattering amplitudes of plant elements and interpreting them as effective polarizabilities. Thus the approximation corresponds to the Foldy-Lax model [11], which is used widely and appropriate for the wavelengths under consideration. Having determined the mean dielectric properties of crowns and short vegetation a 3D grid of points is generated to sample the attenuation function. One grid is required for the incident and scattering directions, and another for the ground reflection direction. For each grid point the broadside ray paths are determined and the attenuation associated with the path determined by finding ray intersections with the boundaries of the understory layer and the crowns. Transmission coefficients are generally close to unity and are not required. In reality of course our crown boundary is completely artificial, as trees do not have well-defined crown volumes, which in itself is good reason for omitting transmission

boundary effects. During the SAR image calculation the grid is used as a look-up table for attenuating scattering amplitudes. A scattering amplitude is assigned the same polarimetric attenuation as that associated with the nearest grid point to the scatterer centre. This provides efficiency and preserves the effects of inhomogeneity.

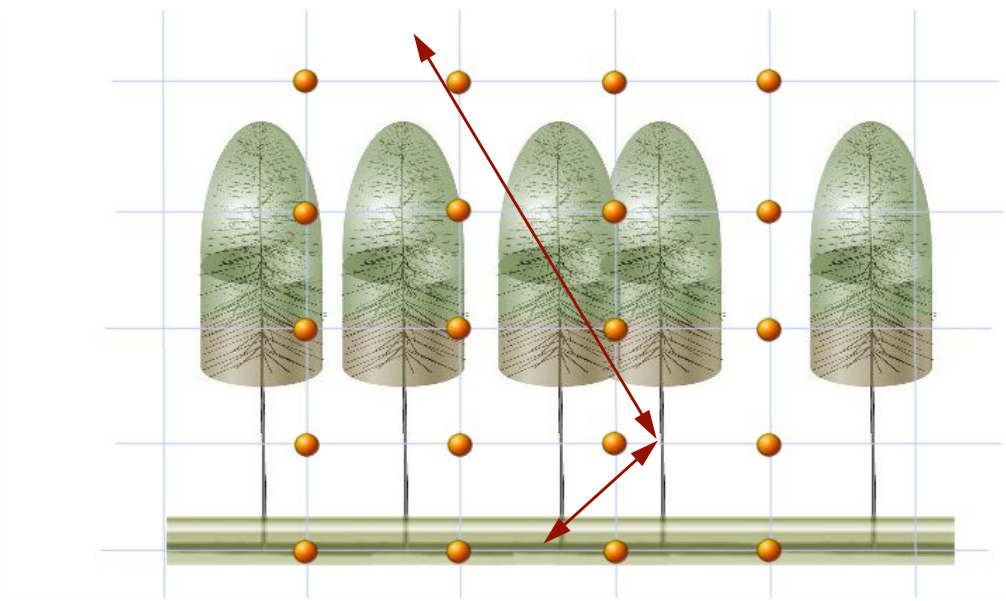

FIGURE 6. Mapping of attenuation into the 3D space occupied by the forest canopy. Direct and reflected ray paths are determined for each grid point. The passage of each ray through crowns and short vegetation is used to determine the attenuation associated with each path

### 6.3. **Direct Ground SAR Scattering.**

The ground surface is divided into flat, rough facets, derived from the large-scale DEM. In principle each facet has a unique realization of surface roughness drawn from an underlying statistical distribution. The polarimetric scattering amplitude of each facet is therefore unique: the variation leading to speckle in the DG term under the right conditions. A fully deterministic approach to the calculation of DG terms is computationally prohibitive. Thus facet scattering amplitudes are calculated using the covariance matrix associated with the underlying surface scattering model calculated in the local frame of the facet. In past work [16] each facet was assigned a complex, polarimetric scattering amplitude drawn from the multivariate Gaussian distribution

$$p\left(\boldsymbol{S}\right) = \frac{\exp\left(-\boldsymbol{S}^{\dagger}\underline{\boldsymbol{C}}^{-1}\boldsymbol{S}\right)}{\pi^{3}\det\underline{\boldsymbol{C}}} \tag{80}$$

sampled using Monte Carlo (MC) techniques. The model employed is two-scale, with the small perturbation model (SPM) used to model the small-scale response. The

copolar channels are perfectly correlated in the SPM, and even with large-scale roughness effects this can lead to unrealistically low polarimetric entropy in the DG term. Thus in practice the covariance matrix approach is adopted using a high (0.95) HH-VV correlation, although this requires careful sampling. Sampling (80) at high HH-VV coherence is problematic, and an alternative approach has been adopted for this study. Since in the SPM HH and VV are perfectly correlated, so we may choose a single, local scattering amplitude factor for each facet, using a single complex Gaussian distribution. To afford a more realistic polarimetric entropy, the local scattering amplitude matrix recovered this way is rotated about the radar line-of-sight by a random angle chosen from a uniform distribution the extent of which is designed to achieve the desired HH-VV coherence in the local frame of the facet [6] (Hajnsek et al., 2003). The final local scattering amplitude is then converted into the radar frame. The hybrid deterministic-stochastic technique is made to preserve interferometric coherence by preserving the random number sequence employed in the MC sampling. In practice any change in incidence angle, however slight, may alter the course of the MC sampling on some facets as the local facet incidence alters the local mean RCS and covariance. This can have a small effect upon coherence for the occasional image pixel. This effect may be minimized by preserving the scaled facet scattering amplitude factors between tracks of the interferometric baseline. Even without this precaution the technique yields high, mean ground coherence as desired.

6.3.1. *Choice of Small-Scale Parameters.*
DG facet scattering amplitudes are calculated in the small perturbation model (SPM) as recorded in [10] (Ogilvy, 1991). Surfaces are taken to be Gaussian correlated and the choice of surface roughness and correlation length are recovered using the users choice of roughness model in the following way. A choice is first made for the surface roughness which complies with the limit if validity of the SPM by assigning

$$\sigma_s = \frac{\alpha}{k\cos\theta} \tag{81}$$

where $\alpha \ll 1$ is a constant of the program. Given a roughness the correlation length corresponding to maximum backscattering coefficient is found as

$$\hat{\lambda}_s = \frac{1}{k\sin\theta} \tag{82}$$

From which is recovered the maximum possible SPM ground brightness at the current incidence, wavelength and roughness, $\hat{\sigma}^0_{hh}$. The default ground brightness is chosen according to frequency, $f$, using $f_p = 0.4333\ GHz$, $f_L = 1.3000\ GHz$, $\delta f = (f - f_p)/(f_L - f_p)$ where $\delta f = 0$ if $f < f_p$ and $\delta f = 1$ if $f > f_L$. Then using $\hat{\sigma}^0_{hhL45} = -28\ dB$ and $\hat{\sigma}^0_{hhP45} = -38\ dB$, $\hat{\sigma}^0_{hh} = \delta f \times \hat{\sigma}^0_{hhL45} + (1 - \delta f) \times \hat{\sigma}^0_{hhP45}$. Now the user defined surface roughness parameter ($0 \le r \le 10$) is used to adjust the desired ground brightness according to $hat\sigma^0_{hh} = hat\sigma^0_{hh} + \Delta^0_{hh} \times (r - 5)/5$, $\Delta^0_{hh} = 3\ dB$. Now if the desired backscatter is greater than the maximum possible, then the correlation length chosen is that corresponding to maximum backscatter, i.e. $\lambda_s = \hat{\lambda}_s$, otherwise

the correlation length is found that yields the desired backscattering coefficient by finding the solution using the Newton-Raphson algorithm.

6.3.2. *The Direct Ground Imaging Algorithm.*
The model for surface scattering is two-scale. The large-scale DEM is divided into triangular facets, with at least 10 per resolution cell to ensure fully-developed speckle in the final image. Each facet has its own orientation and area determined by its location on the large-scale surface, and at the small-scale each facet is presumed to have its own unique realization of roughness, taken from a global, Gaussian roughness model. The facet orientation, area and small-scale roughness are used to determine the scattering amplitude of the surface facet in the global frame. The facet centre position is used to determine the point of focus of the facet in the SAR image plane (the global z = 0 plane), and the interferometric phase of the facet contribution to the DG image. The system point-spread-function (PSF) is used to focus the facet return in the image, and assumes that the facet dimensions are much smaller than resolutions, which they are. We first note that in the Bragg model, or first order small perturbation model (SPM), in the local frame of the facet the cross-polar return is zero and the co-polar returns are perfectly correlated. For fully developed speckle the facet scattering cross sections would display an exponential distribution. We assume that the facets are large compared with the small-scale correlation length so that their scattering cross sections are so distributed. This assumption may be too restrictive in future implementations, however it is sufficient to our purposes at this stage. The local incidence angle for the facet is calculated and used with the dielectric and small-scale properties of the surface to calculate the mean backscattering coefficients, according to [10] (Ogilvy, 1991)

$$\sigma^0_{ab} = 4k^4\sigma_s^2\lambda_s^2\cos^4\theta_i \exp\left(-k^2\lambda_s^2\sin^2\theta_i\right)|B_{ab}|^2 \tag{83}$$

The first-order, small perturbation model scattering coefficients are

$$B_{hh} = \frac{\cos\theta_i - \sqrt{\epsilon_s - \sin^2\theta_i}}{\cos\theta_i + \sqrt{\epsilon_s - \sin^2\theta_i}} \tag{84}$$

$$B_{vv} = \frac{(\epsilon_s - 1)\left(\sin^2\theta_i - \epsilon_s\left(1 + \sin^2\theta_i\right)\right)}{\epsilon_s\cos\theta_i + \sqrt{\epsilon_s - \sin^2\theta}} \tag{85}$$

$$B_{hv} = 0 \tag{86}$$

where $\epsilon_s$ is the soil permittivity, calculated according to [4] (Dobson et al., 1985) for a fixed soil type, and $\theta_i$ is the local incidence angle for the facet. For each facet we then generate a random number, $f$, from an exponential distribution, and a random angle, $\phi$, from a uniform distribution on $[0, 2\pi]$ . The facet scattering amplitude in the local frame of the facet is then determined as

$$\underline{\boldsymbol{S}}_l = \begin{pmatrix} S_{hhl} & S_{hvl} \\ S_{vhl} & S_{vvl} \end{pmatrix} = \begin{pmatrix} \sqrt{fA\sigma^0_{hh}}\exp(i\phi) & 0 \\ 0 & \sqrt{fA\sigma^0_{vv}}\exp(i\phi) \end{pmatrix} \tag{87}$$

where $A$ is the facet area. Now to obtain the scattering matrix in the global frame, we might simply rotate this matrix about the radar line-of-sight (LOS) using the facet orientation. To show how this may be achieved suppose the facet has surface normal vector $\vec{n} = (n_x, n_y, n_z)$, where, in terms of slopes, $s_x$ and $s_y$

$$\gamma_n = \sqrt{1 + s_x^2 + s_y^2} \quad n_z = 1/\gamma_n \quad n_x = -s_x/\gamma_n \quad n_y = -s_y/\gamma_n \tag{88}$$

Now in the global frame the scattering amplitude may be achieved through rotation about the LOS through an angle $\beta_n$ such that

$$\underline{\boldsymbol{S}}_l = \begin{pmatrix} \cos\beta_n & -\sin\beta_n \\ \sin\beta_n & \cos\beta_n \end{pmatrix} \begin{pmatrix} S_{hhl} & 0 \\ 0 & S_{vvl} \end{pmatrix} \begin{pmatrix} \cos\beta_n & -\sin\beta_n \\ \sin\beta_n & \cos\beta_n \end{pmatrix} \tag{89}$$

$$= \begin{pmatrix} \cos\beta_n & -\sin\beta_n \\ \sin\beta_n & \cos\beta_n \end{pmatrix} \begin{pmatrix} S_{hhl}\cos\beta_n & S_{hhl}\sin\beta_n \\ -S_{vvl}\sin\beta_n & S_{vvl}\cos\beta_n \end{pmatrix} \tag{90}$$

$$= \begin{pmatrix} S_{hhl}\cos^2\beta_n + S_{vvl}\sin^2\beta_n & (S_{hhl} - S_{vvl})\cos\beta_n\sin\beta_n \\ (S_{hhl} - S_{vvl})\cos\beta_n\sin\beta_n & S_{hhl}\sin^2\beta_n + S_{vvl}\cos^2\beta_n \end{pmatrix} \tag{91}$$

$$= \begin{pmatrix} S_{hh} & S_{hv} \\ S_{vh} & S_{vv} \end{pmatrix} \tag{92}$$

Expressing the scattering matrix in terms of dyadic products of polarization vectors we have $\underline{\boldsymbol{S}} = S_{hhl}\hat{h}_l\hat{h}_l + S_{vvl}\hat{v}_l\hat{v}_l$ so that

$$S_{hh} = S_{hhl}\left(\hat{h}\cdot\hat{h}_l\right)\left(\hat{h}_l\cdot\hat{h}\right) + S_{vvl}\left(\hat{h}\cdot\hat{v}_l\right)\left(\hat{v}_l\cdot\hat{h}\right) = S_{hhl}\cos^2\beta_n + S_{vvl}\sin^2\beta_n \tag{93}$$

$$S_{hv} = S_{hhl}\left(\hat{h}\cdot\hat{h}_l\right)\left(\hat{h}_l\cdot\hat{v}\right) + S_{vvl}\left(\hat{h}\cdot\hat{v}_l\right)\left(\hat{v}_l\cdot\hat{v}\right) = (S_{hhl} - S_{vvl})\cos\beta_n\sin\beta_n \tag{94}$$

$$S_{vh} = S_{hhl}\left(\hat{v}\cdot\hat{h}_l\right)\left(\hat{h}_l\cdot\hat{h}\right) + S_{vvl}\left(\hat{v}\cdot\hat{v}_l\right)\left(\hat{v}_l\cdot\hat{h}\right) = (S_{hhl} - S_{vvl})\cos\beta_n\sin\beta_n \tag{95}$$

$$S_{vv} = S_{hhl}\left(\hat{v}\cdot\hat{h}_l\right)\left(\hat{h}_l\cdot\hat{v}\right) + S_{vvl}\left(\hat{v}\cdot\hat{v}_l\right)\left(\hat{v}_l\cdot\hat{v}\right) = S_{hhl}\sin^2\beta_n + S_{vvl}\cos^2\beta_n \tag{96}$$

In terms of our rotation model:

$$\left(\hat{h}\cdot\hat{h}_l\right) = +\cos\beta_n \tag{97}$$

$$\left(\hat{h}\cdot\hat{v}_l\right) = -\sin\beta_n \tag{98}$$

$$\left(\hat{v}\cdot\hat{h}_l\right) = +\sin\beta_n \tag{99}$$

$$\left(\hat{v}_l\cdot\hat{v}\right) = +\cos\beta_n \tag{100}$$

By definition we have

$$\hat{h}_l = \frac{\hat{n}\times\hat{k}}{\left|\hat{n}\times\hat{k}\right|} \qquad \hat{v}_l = \hat{h}_l\times\hat{k} \tag{101}$$

In the simulation (and in fact without loss of generality) we write for the incident wave vector, $\hat{k} = (0, \sin\theta, -\cos\theta)$, where $\theta$ is the *global* incidence angle. Substituting our expressions for the facet surface normal vector and the normalized incident wave vector we find (with $\hat{n} = (n_x, n_y, n_z)$)

$$\hat{h} = \frac{\hat{z} \times \hat{k}}{\left|\hat{z} \times \hat{k}\right|} = -\hat{x}\frac{\sin\theta}{|\sin\theta|} = -\hat{x} = (-1, 0, 0) \tag{102}$$

$$\hat{v} = \hat{h} \times \hat{k} = -\hat{y}\cos\theta - \hat{z}\sin theta = (0, \cos\theta, -\sin\theta) \tag{103}$$

$$\hat{h}_l = \frac{\hat{n} \times \hat{k}}{\left|\hat{n} \times \hat{k}\right|} = \frac{-\hat{x}\left(n_y\cos\theta + n_z\sin\theta\right) + \hat{y}n_x\cos\theta + \hat{z}n_x\sin\theta}{\sqrt{n_x^2 + \left(n_y\cos\theta + n_z\sin\theta\right)^2}} \tag{104}$$

$$\hat{v}_l = \hat{h}_l \times \hat{k} = \frac{-\hat{x}n_x - \hat{y}\left(n_y\cos^2\theta + n_z\cos\theta\sin\theta\right) - \hat{z}\left(n_z\sin^2\theta + n_y\cos\theta\sin\theta\right)}{\sqrt{n_x^2 + \left(n_y\cos\theta + n_z\sin\theta\right)^2}} \tag{105}$$

where $\hat{v}_l \cdot \hat{h}_l = 0$. In addition

$$\hat{h} = (-1, 0, 0) \tag{106}$$

$$\hat{v} = (0, -\cos\theta, -\sin\theta) \tag{107}$$

$$\gamma_n = \sqrt{n_x^2 + \left(n_y\cos\theta + n_z\sin\theta\right)^2} \tag{108}$$

$$\hat{h}_l = \left(-n_y\cos\theta - n_z\sin\theta, n_x\cos\theta, n_x\sin\theta\right)/\gamma_n \tag{109}$$

$$\hat{v}_l = \left(-n_x, -n_y\cos^2\theta - n_z\cos\theta\sin\theta, -n_z\sin^2\theta - n_y\cos\theta\sin\theta\right)/\gamma_n \tag{110}$$

Therefore

$$\cos\beta_n = \hat{h} \cdot \hat{h}_l = \left(n_y\cos\theta + n_z\sin\theta\right)/\gamma_n \tag{111}$$

$$\sin\beta_n = -\left(\hat{h} \cdot \hat{v}_l\right) = -n_x/\gamma_n \tag{112}$$

$$\cos\beta_n = \hat{v} \cdot \hat{v}_l = \left(n_y\cos\theta + n_z\sin\theta\right)/\gamma_n \tag{113}$$

$$\sin\beta_n = -\left(\hat{v} \cdot \hat{h}_l\right) = -n_x/\gamma_n \tag{114}$$

Thus [8](Lee et al., 2000).

$$\tan\beta_n = \frac{-n_x}{\left(n_y\cos\theta + n_z\sin\theta\right)} = \frac{s}{\left(\sin\theta - s_y\cos\theta\right)} \tag{115}$$

So to recap, we may find the angle $\beta_n$ from the facet surface normal vector using (115) and form the global scattering matrix from the local scattering matrix using (89) or (93). Now this is perfectly legitimate and for surfaces which are rough on the

large-scale works quite well. However, it is more often the case that surfaces are not very rough on the large-scale, and under these circumstances the polarimetric entropy recovered from the surface term is unrealistically low. To counter this effect we can introduce an additional rotation angle, $\beta$, uniformly distributed between $-[\beta_1, +\beta_1]$ , XBragg [6] (Hajnsek et al., 2003). This helps to increase polarimetric entropy and enhance realism. Thus our final model is in fact XBragg at the small-scale, described by surface roughness, correlation length, soil moisture, and additional tilt angle range. The value of the range of acceptable XBragg rotation angles may be determined by considering the HHVV coherence, which may now be tailored to a desired value by varying $\beta_1$. For a surface with zero slopes, and no large-scale roughness, the XBragg model predicts a coherency matrix

$$k_p = \frac{1}{\sqrt{2}}\left[S_{hh} + S_{vv},\ S_{hh} - S_{vv},\ 2S_{hv}\right]^T \tag{116}$$

$$S_{hh} = S_{hhl}\left(\hat{h}\cdot\hat{h}_l\right)\left(\hat{h}_l\cdot\hat{h}\right) + S_{vvl}\left(\hat{h}\cdot\hat{v}_l\right)\left(\hat{v}_l\cdot\hat{h}\right) \quad = S_{hhl}\cos^2\beta_n + S_{vvl}\sin^2\beta_n \tag{117}$$

$$S_{hv} = S_{hhl}\left(\hat{h}\cdot\hat{h}_l\right)\left(\hat{h}_l\cdot\hat{v}\right) + S_{vvl}\left(\hat{h}\cdot\hat{v}_l\right)\left(\hat{v}_l\cdot\hat{v}\right) \quad = \left(S_{hhl} - S_{vvl}\right)\cos\beta_n\sin\beta_n \tag{118}$$

$$S_{vh} = S_{hhl}\left(\hat{v}\cdot\hat{h}_l\right)\left(\hat{h}_l\cdot\hat{h}\right) + S_{vvl}\left(\hat{v}\cdot\hat{v}_l\right)\left(\hat{v}_l\cdot\hat{h}\right) \quad = \left(S_{hhl} - S_{vvl}\right)\cos\beta_n\sin\beta_n \tag{119}$$

$$S_{vv} = S_{hhl}\left(\hat{v}\cdot\hat{h}_l\right)\left(\hat{h}_l\cdot\hat{v}\right) + S_{vvl}\left(\hat{v}\cdot\hat{v}_l\right)\left(\hat{v}_l\cdot\hat{v}\right) \quad = S_{hhl}\sin^2\beta_n + S_{vvl}\cos^2\beta_n \tag{120}$$

$$\begin{aligned} k_p &= \frac{1}{\sqrt{2}}\begin{bmatrix} S_{hhl} + S_{vvl} \\ \left(S_{hhl} - S_{vvl}\right)\left(\cos^2\beta_n - \sin^2\right) \\ 2\left(S_{hhl} - S_{vvl}\right)\cos\beta\sin\beta \end{bmatrix} \\ &= \frac{1}{\sqrt{2}}\left[S_{hhl} + S_{vvl},\ \left(S_{hhl} - S_{vvl}\right)\cos 2\beta_n,\ 2\left(S_{hhl} - S_{vvl}\right)\sin 2\beta\right]^T \end{aligned} \tag{121}$$

with

$$2T_{11} = \langle |S_{hhl} + S_{vvl}|^2\rangle \tag{122}$$

$$2T_{12} = \langle \left(S_{hhl} + S_{vvl}\right)\left(S_{hhl} + S_{vvl}\right)^*\rangle\langle\cos 2\beta\rangle \tag{123}$$

$$2T_{22} = \langle |S_{hhl} - S_{vvl}|^2\rangle\langle\cos^2 2\beta\rangle \tag{124}$$

$$2T_{33} = \langle |S_{hhl} - S_{vvl}|^2\rangle\langle\sin^2 2\beta\rangle \tag{125}$$

Where in the XBragg model

$$\langle\cos 2\beta\rangle = \mathrm{sinc}\left(2\beta_1\right) \tag{126}$$

$$\langle\cos^2 2\beta\rangle = \frac{1}{2}\left[1 + \mathrm{sinc}\left(4\beta_1\right)\right] \tag{127}$$

$$\langle\sin^2 2\beta\rangle = \frac{1}{2}\left[1 - \mathrm{sinc}\left(4\beta_1\right)\right] \tag{128}$$

Whereas for a surface with non-zero slopes, but zero XBragg rotation range we might have

$$2T_{11} = \langle |S_{hhl} + S_{vvl}|^2 \rangle \tag{129}$$

$$2T_{12} = \langle (S_{hhl} + S_{vvl})\,(S_{hhl} + S_{vvl})^* \rangle \cos 2\beta_0 \tag{130}$$

$$2T_{22} = \langle |S_{hhl} - S_{vvl}|^2 \rangle \cos^2 2\beta_0 \tag{131}$$

$$2T_{33} = \langle |S_{hhl} - S_{vvl}|^2 \rangle \sin^2 2\beta_0 \tag{132}$$

For a more general case, with rotation angle $\beta = \beta_{XBragg} + \beta_{facet}$ we may examine the mean co-polar powers and co-polar coherency

$$\langle |S_{hh}|^2 \rangle = \langle |S_{hhl}|^2 \rangle \langle \cos^4 \beta \rangle + \langle |S_{vvl}|^2 \rangle \langle \sin^4 \beta \rangle + 2\mathbb{R}\left\{ \langle S_{hhl} S_{vvl}^* \rangle \right\} \langle \cos^2 \beta \sin^2 \beta \rangle \tag{133}$$

$$\langle |S_{vv}|^2 \rangle = \langle |S_{hhl}|^2 \rangle \langle \cos^4 \beta \rangle + \langle |S_{hhl}|^2 \rangle \langle \sin^4 \beta \rangle + 2\mathbb{R}\left\{ \langle S_{hhl} S_{vvl}^* \rangle \right\} \langle \cos^2 \beta \sin^2 \beta \rangle \tag{134}$$

$$\langle S_{hh} S_{vv}^* \rangle = \left( \langle |S_{hhl}|^2 \rangle + \langle |S_{vvl}|^2 \rangle \right) \langle \cos^2 \beta \sin^2 \beta \rangle + \langle S_{hhl} S_{vvl}^* \rangle \langle \cos^4 \beta \rangle + \langle S_{hhl}^* S_{vvl} \rangle \langle \sin^4 \beta \rangle \tag{135}$$

Since in the Bragg model HH and VV are perfectly correlated so $\langle S_{hhl} S_{vvl}^* \rangle$ is real and these simplify to

$$\langle |S_{hh}|^2 \rangle = \langle |S_{hhl}|^2 \rangle \langle \cos^4 \beta \rangle + \langle |S_{vvl}|^2 \rangle \langle \sin^4 \beta \rangle + 2\langle S_{hhl} S_{vvl}^* \rangle \langle \cos^2 \beta \sin^2 \beta \rangle \tag{136}$$

$$\langle |S_{vv}|^2 \rangle = \langle |S_{hhl}|^2 \rangle \langle \cos^4 \beta \rangle + \langle |S_{hhl}|^2 \rangle \langle \sin^4 \beta \rangle + 2\langle S_{hhl} S_{vvl}^* \rangle \langle \cos^2 \beta \sin^2 \beta \rangle \tag{137}$$

$$\langle S_{hh} S_{vv}^* \rangle = \left( \langle |S_{hhl}|^2 \rangle + \langle |S_{vvl}|^2 \rangle \right) \langle \cos^2 \beta \sin^2 \beta \rangle + \langle S_{hhl} S_{vvl}^* \rangle \left( \langle \cos^4 \beta + \sin^4 \beta \rangle \right) \tag{138}$$

$$\langle |S_{hv}|^2 \rangle = \langle |S_{hhl} - S_{vvl}|^2 \rangle \langle \cos^2 \beta \sin^2 \beta \rangle \tag{139}$$

Note that a value of $\beta_1 = \pi/3$ results in a reduced HHVV coherence of 0.94. Note also that

$$\langle \cos^2 \beta \rangle = \frac{1}{2}\left[1 + \mathrm{sinc}\,(2\beta_1)\right] \tag{140}$$

$$\langle \sin^2 \beta \rangle = \frac{1}{2}\left[1 - \mathrm{sinc}\,(2\beta_1)\right] \tag{141}$$

$$\langle \cos^2 \beta \sin^2 \beta \rangle = \frac{1}{4}\left[\frac{1}{2}\left(1 + \mathrm{sinc}\,(2\beta_1)\right) - \cos^3 \beta_1 \mathrm{sinc} \beta_1\right] \tag{142}$$

and

$$\langle \cos^4 \beta \rangle = \langle \cos^2 \beta \rangle - \langle \cos^2 \beta \sin^2 \beta \rangle \tag{143}$$

$$\langle \sin^4 \beta \rangle = \langle \sin^2 \beta \rangle - \langle \cos^2 \beta \sin^2 \beta \rangle. \tag{144}$$

6.3.3. *XBragg on a Tilted Surface.*
Our results so far have been quoted for a uniform, symmetric distribution of XBragg rotation angle. However if the large-scale roughness is zero (i.e. the surface is flat, but rough on the small-scale), but the mean surface has non-zero slopes, then the distribution of rotation angles is offset by an amount determined by the mean slope. Let the mean rotation angle satisfy

$$\tan\beta_0 = \frac{s}{(\sin\theta - s_y\cos\theta)} \tag{145}$$

where the slopes are those of the mean surface. Our equations for the mean backscattering coefficients and correlation, (133 - 139), still hold but the angular brackets denote averaging over the offset distribution. For example we now seek

$$\begin{aligned}\langle\cos^2(\beta+\beta_0)\rangle &= \langle(\cos\beta\cos\beta_0 - \sin\beta\sin\beta_0)^2\rangle \\ &= \langle\cos^2\beta\rangle\cos^2\beta_0 - \langle\sin^2\beta\rangle\sin^2\beta_0 \\ &= \frac{\cos^2\beta_0}{2}\left[1+\operatorname{sinc}(2\beta_1)\right] - \frac{\sin^2\beta_0}{2}\left[1-\operatorname{sinc}(2\beta_1)\right]\end{aligned} \tag{146}$$

$$\begin{aligned}\langle\sin^2(\beta+\beta_0)\rangle &= \langle(\sin\beta\cos\beta_0 + \cos\beta\sin\beta_0)^2\rangle \\ &= \langle\sin^2\beta\rangle\cos^2\beta_0 + \langle\sin^2\beta\rangle\sin^2\beta_0 \\ &= \frac{\cos^2\beta_0}{2}\left[1-\operatorname{sinc}(2\beta_1)\right] + \frac{\sin^2\beta_0}{2}\left[1+\operatorname{sinc}(2\beta_1)\right]\end{aligned} \tag{147}$$

$$\begin{aligned}\langle\cos^2(\beta+\beta_0)\sin^2(\beta+\beta_0)\rangle &= \langle\cos^2\beta\sin^2\beta\rangle\left(\cos^4\beta_0+\sin^4\beta_0-4\cos^2\beta_0\sin^2\beta_0\right) \\ &\quad + \left(\langle\cos^4\beta\rangle+\langle\sin^4\beta\rangle\right)\cos^2\beta_0\sin^2\beta_0\end{aligned} \tag{148}$$

These results may be used to estimate the direct ground backscattering coefficients using equations (133) and (139).

6.3.4. *DG Algorithm Summary.*
The DEM is broken into facets and for each facet a speckle phase is assigned from a uniformly random distribution on $[0, 2\pi]$, and a scattering power scaling is generated from an exponential distribution. The mean local scattering powers for the facet are calculated in the SPM using the facet area and orientation, and the local scattering amplitude matrix for the facet is determined from (87). The facet orientation is used to define a LOS rotation angle from (115), and to this is added a random X-Bragg rotation. The global frame scattering amplitude is determined from (93) using the combined rotation angle. Attenuation effects are then incorporated into the scattering amplitude using the lookup table, before focusing the contribution in the SAR image.

6.4. **Direct-Vegetation (DV) and Direct-Forest (DF) SAR Scattering.**
The strategies employed to calculate both the short vegetation and forest contributions are very similar, differing in the main by the scattering models used for plant elements based on dimensions relative to wavelength. Another subtle difference is in the way the totality of the contributions from different classes of scatterer is estimated.

6.4.1. *Deterministic Model.*
Volume terms for large scatterers, i.e. stems, primary and secondary branches are calculated in a deterministic fashion. Branches are divided into short segments, and scattering is modeled using an approximate but efficient form for the truncated infinite cylinder approximation [2, 7] (Fung, 1994, Karam et al., 1988).

6.4.2. *Hybrid Stochastic Model.*
The numerous small plant elements in each crown and in the short vegetation layer are treated efficiently by using a hybrid approach. When calculating scattering terms the understory is divided into areas smaller than a resolution cell, and each is populated with a number of random realizations of occupying scattering elements. Crowns are treated in a similar fashion: distributing realizations throughout the crown evenly. Scattering amplitudes are scaled if the number of plant elements that would actually be found in each area or volume is greater than the number of realizations. Realizations are preserved between calculations for each track on the interferometric baseline: in this fashion computational efficiency is maintained whilst preserving both backscattering coefficient, and coherence behavior with baseline. Scattering by small elements is calculated using the Rayleigh-Gans approach [17] (Williams and Quegan, 1997), whilst the scattering due to very small objects, such as Pine needles, at low frequency is, if not significant, ignored. These considerations serve to make the calculation computationally efficient. Vegetation dielectric permittivities are calculated according to [13] (Ulaby et al., 1987), using predetermined plant element water contents.

6.5. **Surface-Vegetation (SV) and Surface-Forest (SF) SAR Scattering.**
The calculation of the "effective scattering centre" [16] (Williams, 2006) is a heavy computational burden if we require account to be taken of local slope. Thus the approximation is adopted that the reflection is from the mean terrain surface, and is modulated by the mean roughness. Thus the 'effective scattering centre' is found readily for all scatterers as the projection onto the mean terrain surface of the scattering centre [16]. Both the deterministic and hybrid stochastic approaches are adopted for the SV calculation in the manners previously described in the DV section. The ground-surface-volume-ground-surface (SVS) interaction is not considered as it is generally found not to be significant. Note that whilst DV terms display layover, the same is not true for SV terms. If included SVS terms would display layover away from the radar, and appear to have phase centers below the soil surface. In the shadow region of the forest stand the SVS term may thus appear dominant and lead to confusing tree height inversions. To avoid this, and the associated high costs of its computation, the SVS term is omitted.

6.6. **The Point Spread Function.**
The complex scattered returns rom each scattering element are then added to the corresponding image pixels. However, in reality the band-limited nature of SAR systems causes the scattered energy to spill over into neighboring pixels. The amount and extent of this energy spread is a function of SAR system parameters such as bandwidth, sampling and weighting functions, encapsulated in the system point spread function. **PolSARproSim**$^{+}$simulates a point spread function that allows more flexibility in simulating different instrument parameters. For a scattered field $S(x,y)$, the amount of scattered energy spilled over to a neighboring pixel, say $S(\tilde{x},\tilde{y})$, is given by

$$S(\tilde{x},\tilde{y}) = \left( \int_{\tilde{x}-N\delta x/2}^{\tilde{x}+N\delta x/2} I(\delta x')\, d\delta x' \int_{\tilde{y}-M\delta y/2}^{\tilde{y}+M\delta y/2} I(\delta g')\, d\delta g' \right) S(x,y) \tag{149}$$

where $\delta x$ and $\delta y$ are the azimuth and ground-range pixel dimensions respectively, while $N$ and $M$ set the programmer defined integration limits in number of pixels. The point spread functions in azimuth and range, $I(\delta x)$ and $I(\delta y)$ respectively, are given by

$$I(\delta_x) = \operatorname{sinc}\left(\frac{\pi\delta_x}{\Delta x}\right) + \frac{1}{\pi}\frac{1-\eta}{1+\eta}\sin\left(\frac{\pi\delta_x}{\Delta x}\right)\left(\frac{\delta_x/\Delta x}{1-(\delta_x/\Delta x)^2}\right) \tag{150}$$

$$I(\delta_y) = \operatorname{sinc}\left(\frac{\pi\delta_y\sin\theta}{\Delta y_s}\right) + \frac{1}{\pi}\frac{1-\eta}{1+\eta}\sin\left(\frac{\pi\delta_y\sin\theta}{\Delta y_s}\right)\left(\frac{\delta_y\sin\theta/\Delta y_s}{1-(\delta_y\sin\theta/\Delta y_s)^2}\right) \tag{151}$$

where $\theta$ is the incidence angle, $\Delta x$, and $\Delta y_s$ the azimuth and slant range resolutions respectively while $\eta$, $(0 \leq \eta \leq 1)$, the user controlled beam-broadening factor, defines the shape of the weighting function such that at $\eta = 1$ there is no beam broadening, i.e. the point spread function reduces to a Sinc, while $\eta = 0.0$ leads to a Hanning weighting function etc. Figure 7 plots simulated range spectra for a system with slant range

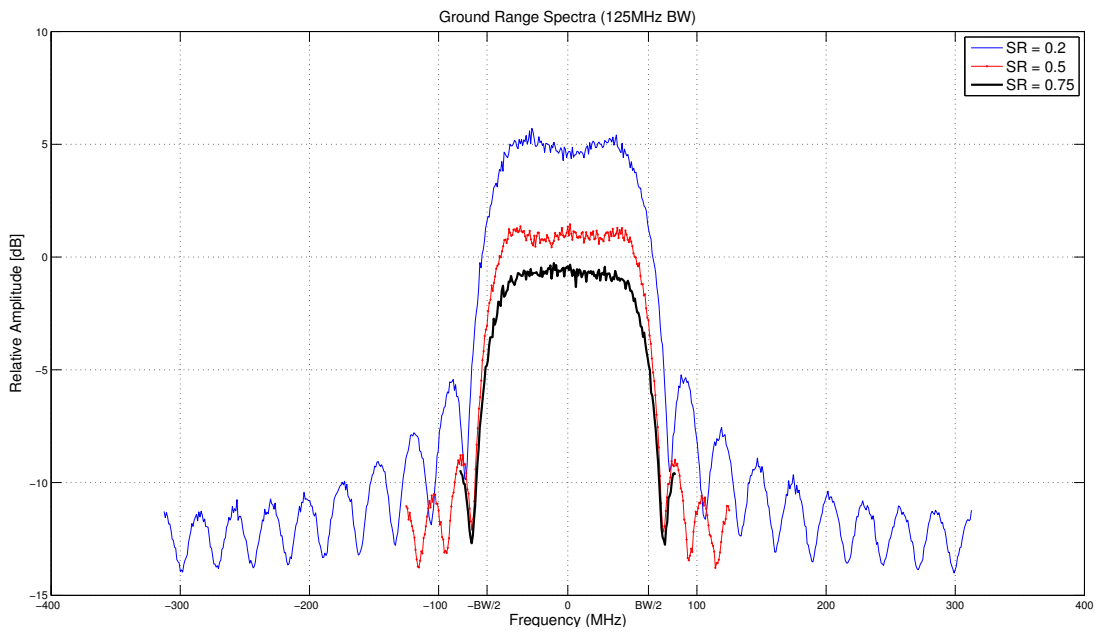


FIGURE 7. Simulated range spectra for different range sampling factors (SR = 0.2, 0.5 and 0.75)

resolution of 1.2 m (approximately 125 MHz bandwidth) for three different sampling rates that can be controlled using the parameter '`gnd_range_sampling_factor`' in the input file. The three plots are for sampling factors of 0.2, 0.5 and 0.75, where for instance the sampling frequency when SR = 0.2 is 625 MHz, see (18) for details. Azimuth spectra can be similarly controlled using '`azimuth_sampling_factor`'. Figure 8 plots the range spectra for three different beam broadening factors, at $\eta = 1.0$, 0.5 and 0.04 with $\eta = 1.0$ corresponding to a Sinc weighting function and $\eta = 0.04$ to Hamming weighting. Both the range and azimuth spectra are shaped by the same parameter specified in the input file as '`PSF_broadening_factor`'. Figure 9 shows

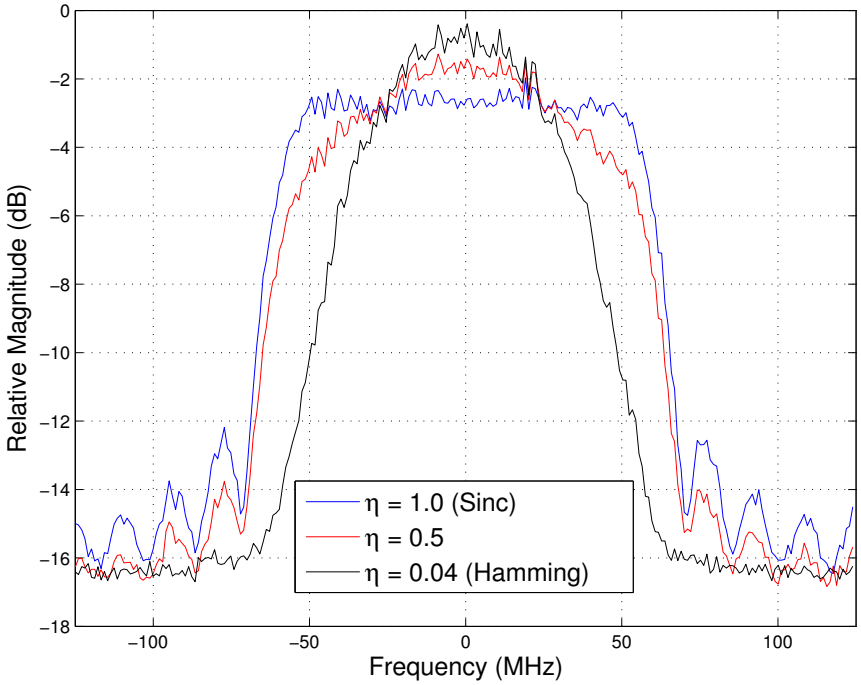


FIGURE 8. Simulated range spectra for different beam broadening factors ($\eta$)

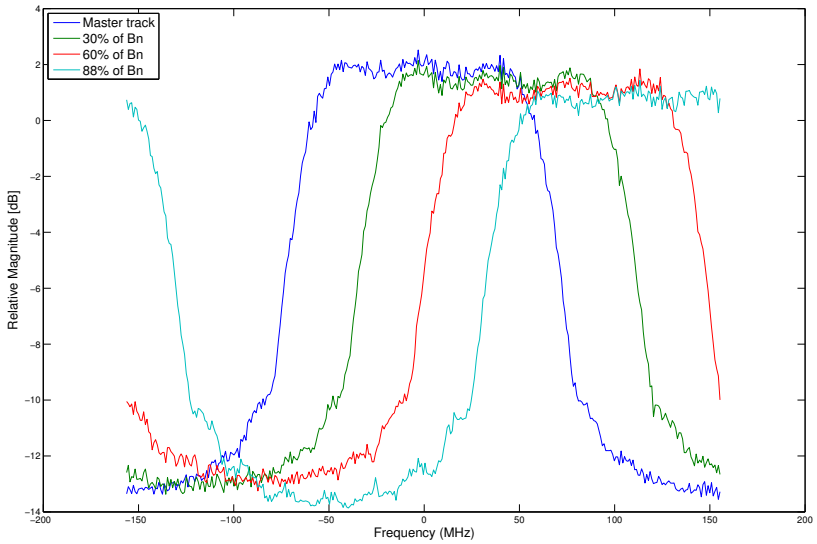


FIGURE 9. Simulated range spectra for different baselines chosen as fractions of the critical baseline, Bn. The shift in range spectra seen here is due to varying geometry, as expected.

range spectra for scenes simulated with varying baselines chosen as a function of the critical baseline (i.e. the baseline where geometric decorrelation is at a maximum). The observed spectral shift corresponds to the predicted Prati shift [5]. Figure 10 shows the corresponding coherence for interferograms formed using these scenes. The amount of decorrelation follows the geometric decorrelation model given by

$$\gamma_G = 1 - \frac{B_\perp}{B_{\perp_c}} \tag{152}$$

where the $B_\perp$ is the perpendicular baseline and the critical perpendicular baseline, $B_{\perp_c}$ is given by

$$B_{\perp_c} = \frac{\lambda R \tan(\theta - \alpha)}{2\Delta y_s} \tag{153}$$

where $R$ is the slant range to the target, $\theta$ is the incidence angle, $\alpha$ is the local terrain slope (in this case it was zero), and $\Delta y_s$ is the slant range resolution.

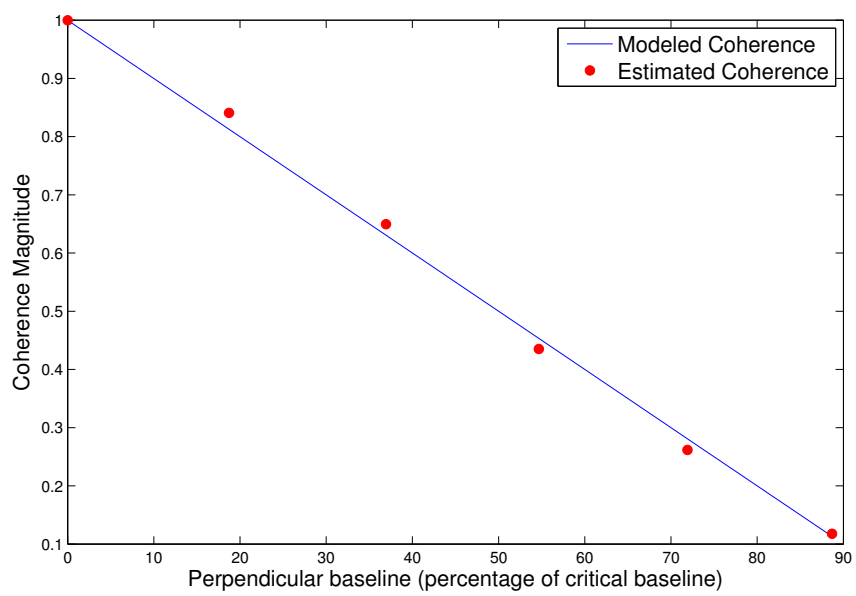


FIGURE 10. Simulated coherence for a stack of interferograms with baselines chosen as percentages of the critical baseline. The trend in simulated coherence follows what is expected from models of baseline decorrelation, $\gamma_G$.

### 6.7. **Thermal Noise.**

The amount of thermal noise in a SAR image is a function of many system and imaging parameters such as system gain, losses, incidence angle etc. While useful, it is too complex simulate all such factors here, however **PolSARproSim**$^+$does allow the simulation of thermal noise through the parameter '`NESZ`', the noise equivalent sigma

nought for a SAR measurement that is supplied in dB for each polarization channel. The simulated scattered field with thermal noise, $\tilde{S}_{pq}$, is then given by

$$\tilde{S}_{pq} = S_{pq} + N_{pq} \tag{154}$$

$$N_{pq} = n_{x_{pq}} + i n_{y_{pq}} \tag{155}$$

$$n_{x_{pq}}, n_{y_{pq}} \sim N\left(0, \frac{\sigma^0_{n_{pq}}}{\sqrt{2}}\right) \tag{156}$$

$$\sigma^0_{n_{pq}} = 10^{(-\mathrm{NESZ}_{pq}/20)} \tag{157}$$

where $S_{pq}$ is the simulated scattered field for a polarimetric channel $pq = [\mathrm{HH, HV, VH, VV}]$, $N_{pq}$ is the simulated thermal noise specified through the noise equivalent sigma nought, $\mathrm{NESZ}_{pq}$ in dB units.

### 6.8. **Temporal Decorrelation.**

Interferometric measurements from repeat-pass instruments are sensitive to changes in the properties of scattering objects between multiple acquisitions. The primary drivers of temporal decorrelation are change in scatterer positions due to wind, growth or anthropogenic activities, and change in electrical characteristics caused by change in moisture. **PolSARproSim**$^+$can simulate the effects of temporal decorrelation by randomly moving or changing the moisture of individual scattering elements such as branches leaves etc. The amount of change in moisture or position is specified for each track as a quadratic or exponential function of the height of the scatterer (relative to the ground or the canopy). The position of a scattering element (a three dimensional vector from the scene center to the center of the scatterer) is changed such that

$$\vec{p}_{new} = \vec{p} + d\vec{p} \tag{158}$$

where $\vec{p}_{new}$ is the new position of the scatterer (originally at $\vec{p}$), while the change in position, $d\vec{p}$ is given by

$$d\vec{p} = \begin{bmatrix} dx \\ dy \\ dx \end{bmatrix} \tag{159}$$

where $dx$, $dy$, $dz$ are each Normally distributed random variables such that,

$$dx,\, dy,\, dz \sim N\left(0, \sigma\left(h\right)\right) \tag{160}$$

where $\sigma\left(h\right)$ is specified for each track as a set of coefficients ($A$, $B$, and $C$) in the input parameter file using the parameter `position_change_coeff_track_N` for each track `N`, in case of a quadratic function as

$$\sigma\left(h\right) = Ah^2 + Bh + C \tag{161}$$

or for an exponential function as

$$\sigma\left(h\right) = Ae^{Bh} + C. \tag{162}$$

Change in moisture is simulated as a mean offset to the original scatterer moisture, $M$, such that

$$M_{new} = M + dM \tag{163}$$

where $M_{new}$ is the new moisture value for a scatterer, while the change $dM$ is modeled as a Normally distributed random variable such that

$$dM \sim N\left(m\left(h\right), m\left(h\right)/10\right) \tag{164}$$

where $m\left(h\right)$ is specified for each track as a set of coefficients ($A$, $B$, and $C$) in the input parameter file using the parameter `moisture_change_coeff_track_N` for each track `N`, in case of a quadratic function as

$$m\left(h\right) = Ah^2 + Bh + C \tag{165}$$

or for an exponential function as

$$m\left(h\right) = Ae^{Bh} + C. \tag{166}$$

## 7. File Formats and Calling Convention

There are a number of files required for the correct operation of **PolSARproSim**$^+$. These include

- Parameter file (mandatory)
- Forest definition file (optional)
- Species allometry file (mandatory)
- External DEM file (optional)

This section describes the formats and individual parameters for these files.

### 7.1. PolSARproSim$^+$Calling Convention.

**PolSARproSim**$^+$is called from the command line with three arguments; the full path and filename for the input parameter file, the prefix for the output files and the location for the output files. For example

```
[dave@hal /home/workdir/]$ polsarprosimplus input.sar run slcs/
```

That is to say the executable is called with name '`polsarprosimplus`' and is passed three string variables on the command line. The first variable is the input parameter file including the directory path, the second is the prefix for the output files while the third is the pathname of the directory where **PolSARproSim**$^+$is to write output files. In this particular case **PolSARproSim**$^+$is called with the with the prefix '`run`' and the input file `input.sar` presumed to be in `/home/workdir`. All the output files in this case will be stored in `/home/workdir/slcs/`, e.g. the master (track 0) HH image file will be called `/home/workdir/slcs/run_track000_HH.slc` Aside from the polarimetric SAR imagery, which is output in binary form (formats are discussed later), other forms of output include a graphic file with a simplified representation of the simulated forest stand as viewed from the platform at broadside, a large-scale ground height map, and attenuation mapping data. In addition the simulation writes three text files in the

output directory: prefix_call.txt, prefix.out and prefix.log. The first of these simply details the call made to **PolSARproSim**$^+$, the second reproduces the contents of the input file, and lists the details of the SAR images produced. The log file contains detailed information about the performance of **PolSARproSim**$^+$and is maintained for debugging and trouble-shooting purposes, and is not intended for the user.

### 7.2. **Input Parameter File.**

The input parameter file is an ASCII text file with variables needed to set up a particular **PolSARproSim**$^+$simulation run. Figure 11 shows an example parameter file. Each line of the input file consists of a keyword with a corresponding initial value (one per line) followed by optional comments describing the parameter associated with the value. All comments need to be preceded by the '#' symbol, and the keywords should not be repeated elsewhere in the file. The parameters can be listed in any order in the input file, although they are grouped together for convenience. A short description of each parameter follows.

- **randn_seed**: This is an integer value used to seed the random number generator.
- **fast_mode:** This integer flag (either 0 or 1) is used to enable or disable the fast mode approximation in **PolSARproSim**$^+$. This fast mode approximation assumes that the scattering matrix does not change between tracks, and that the scatterers only move relative to the radar, changing their phase and scattering center. This approximation saves computation time, hence referred to as fast mode. If the electrical properties of scatterers are changed for a track this mode is turned off for that particular track only.

The following set of parameters are used to determine radar viewing geometry.

- **tracks**: The number of requested tracks. This is an integer that can have any value above 1. Since memory is allocated for each track simultaneously, requesting too many tracks might slow down the simulator.
- **center_slant_range_master**: The slant range, in meters, to the center of the scene for the master track. The floating point value must only be larger than zero.
- **incidence_angle_master**: The incidence angle, in degrees, for the center of the scene for the master track. The floating point value must be larger than zero and less than 90 degrees.
- **baseline_track_n**: This parameter defines the baseline for each track relative to the geometry of the master track. This is a two dimensional floating point vector, separated by a tab or a space, where the first entry is the horizontal component while the second entry is the vertical component of the physical baseline, defined in meters. To differentiate tracks, the number of each track is appended to the keyword baseline_track, for instance baseline_track_1 refers to the baseline vector for the first slave track and so on.

The following set of parameters describe radar instrument parameters

- **radar_frequency**: The centre frequency of the radar as type double and is given in Gigahertz.

```
# PolSARproSim+ Configuration File
# - All comments should start with the '#' symbol, filenames should not have the '#' symbol in them
# - parameter names should not be repeated elsewhere in the file

#-----------------------------------------------------------------------------------------------------------------------------------------------------------#
# PARAMETER NAME                VALUE                   DESCRIPTION                                                                                         #
#-----------------------------------------------------------------------------------------------------------------------------------------------------------#
randn_seed                      45233                   # Random number generator seed                                                                      #
fast_mode                       0                       # 1: Enable Fast Mode approx.,  0: recompute every scattering matrix                                 #

#-----------------------------------------------------------------------------------------------------------------------------------------------------------#
# Parameters for defining radar viewing geometry                                                                                                            #
#-----------------------------------------------------------------------------------------------------------------------------------------------------------#
tracks                          2                       # The number of requested tracks (including master track)                                           #
center_slant_range_master       14433.756730            # Slant range (broadside platform to scene centre) in meters for master track                       #
incidence_angle_master          40.0000                 # Incidence angle for master track in degrees                                                       #
baseline_track_1                100.0   50.00           # horizontal and vertical baseline components for slave track 1 [m,m]                               #
baseline_track_2                0.0     0.00            # horizontal and vertical baseline components for slave track 2 [m,m]                               #
baseline_track_3                0.0     0.00            # horizontal and vertical baseline components for slave track 3 [m,m]                               #
baseline_track_4                40.0    0.00            # horizontal and vertical baseline components for slave track 4 [m,m]                               #
baseline_track_5                50.0    0.00            # horizontal and vertical baseline components for slave track 5 [m,m]                               #
baseline_track_6                0.0     0.00            # horizontal and vertical baseline components for slave track 6 [m,m]                               #
baseline_track_7                4000.0  0.00            # horizontal and vertical baseline components for slave track 7 [m,m]                               #

#-----------------------------------------------------------------------------------------------------------------------------------------------------------#
# Radar system parameters                                                                                                                                   #
#-----------------------------------------------------------------------------------------------------------------------------------------------------------#
radar_frequency                 1.25                    # Center frequency in GHz                                                                           #
azimuth_resolution              1.2                     # Azimuth resolution (width at 3-dB point) in meters                                                #
slant_range_resolution          1.2                     # Slant range resolution (width at 3-dB point) in meters                                            #
PSF_broadening_factor           1.00                    # Factor (>0 and <1) that determines PSF shape: 0.0 Hann, 0.08 Hamming, 1.0 Uniform                 #
azimuth_sampling_factor         0.75                    # Ratio of pixel dimension to resolution in azimuth, default 2/3                                   #
gnd_range_sampling_factor       0.75                    # Ratio of pixel dimension to resolution in ground range, default 2/3                              #
NESZ                            -42.0 -45.0 -45.0 -45.0 # System noise equivalent sigma nought for each polarization (HH, HV, VH, VV) [in dB]               #

#-----------------------------------------------------------------------------------------------------------------------------------------------------------#
# Surface parameters                                                                                                                                        #
#-----------------------------------------------------------------------------------------------------------------------------------------------------------#
input_dem                       0                       # 1: Input DEM from external binary file 0: generate DEM from slope parameters                      #
external_DEM_file               mydem.bin               # External DEM file name                                                                            #
DEM_azimuth_pixels              175                     # Number of pixels in azimuth (or x-) direction of external DEM (unitless)                          #
DEM_range_pixels                209                     # Number of pixels in ground range (or y-) direction of external DEM (unitless)                     #
DEM_azimuth_resolution          1.48571                 # Resolution of external DEM in azimuth [m]                                                         #
DEM_range_resolution            1.48148                 # Resolution of external DEM in ground range [m]                                                    #
ground_slope_azimuth            0.000                   # Ground slope in azimuth direction (dimensionless) (not used with external DEM)                    #
ground_slope_range              0.0000                  # Ground slope in ground range direction (dimensionless) (not used with external DEM)               #
ground_moisture                 0 0 0 6 8 10            # Ground moisture content for each track: 0 = driest ... 10 = wettest (default = 0)                 #
roughness_model                 10                      # surface roughness model: 0 = perfectly smooth, 10 = very rough                                    #

#-----------------------------------------------------------------------------------------------------------------------------------------------------------#
# Forest parameters                                                                                                                                         #
#-----------------------------------------------------------------------------------------------------------------------------------------------------------#
input_forest                    1                       # 1: Input forest from forest xml file 0: generate forest of global tree species                    #
forest_file                     forest.xml              # Input forest XML file                                                                             #
allometry_file                  allometry.xml           # Input Allometry file                                                                              #
global_tree_species             ACRU                    # Tree species: this should be matched with a species tag in allometry file                         #
global_tree_height              30.5000                 # Mean tree height of internally generated stand in meters                                          #
forest_stand_area               10000.0000              # Area of the internally generated forest stand in square meters                                    #
stem_density                    30.0000                 # Desired stand density of the internally generated stand in stems per hectare                      #
forest_output                   Forest_output.xml       # Output forest XML file used if internally generated forest option enabled                         #
draw_forest                     0                       # 1: Draw an image of the forest (NOT RECOMMENDED FOR THIS VER),  0: skip the drawing               #

#-----------------------------------------------------------------------------------------------------------------------------------------------------------#
# Parameters for modeling temporal decorrelation                                                                                                            #
#-----------------------------------------------------------------------------------------------------------------------------------------------------------#
change_height_reference         1                       # 0: Use ground as height reference, 1: Use canopy bottom as height reference                       #
# -- track dependent parameters for changing scatterer position                                                                                             #
position_change_model_track_1   1                       # 0: No change, 1: Polynomial, 2: Exponential                                                       #
position_change_coeff_track_1   0.0 0.004 0.00          # Position Change Model Coefficients: A, B, C (Ax^2+Bx+C, Ae^(Bx)+C) for slave track 1              #
position_change_model_track_2   1                       # 0: No change, 1: Polynomial, 2: Exponential                                                       #
position_change_coeff_track_2   0.00004 0.0 0.00        # Position Change Model Coefficients: A, B, C (Ax^2+Bx+C, Ae^(Bx)+C) for slave track 2              #
position_change_model_track_3   1                       # 0: No change, 1: Polynomial, 2: Exponential                                                       #
position_change_coeff_track_3   0.00008 0.0 0.0         # Position Change Model Coefficients: A, B, C (Ax^2+Bx+C, Ae^(Bx)+C) for slave track 3              #
position_change_model_track_4   1                       # 0: No change, 1: Polynomial, 2: Exponential                                                       #
position_change_coeff_track_4   0.00012 0.0 0.0         # Position Change Model Coefficients: A, B, C (Ax^2+Bx+C, Ae^(Bx)+C) for slave track 4              #
position_change_model_track_5   1                       # 0: No change, 1: Polynomial, 2: Exponential                                                       #
position_change_coeff_track_5   0.00016 0.0 0.0         # Position Change Model Coefficients: A, B, C (Ax^2+Bx+C, Ae^(Bx)+C) for slave track 5              #
position_change_model_track_6   1                       # 0: No change, 1: Polynomial, 2: Exponential                                                       #
position_change_coeff_track_6   0.00020 0.0 0.0         # Position Change Model Coefficients: A, B, C (Ax^2+Bx+C, Ae^(Bx)+C) for slave track 6              #
# -- track dependent parameters for changing scatterer moisture                                                                                             #
moisture_change_model_track_1   1                       # 0: No change, 1: Polynomial, 2: Exponential                                                       #
moisture_change_coeff_track_1   0.0 0.01 0.00           # Moisture Change Model Coefficients: A, B, C (Ax^2+Bx+C, Ae^(Bx)+C) for slave track 1              #
moisture_change_model_track_2   1                       # 0: No change, 1: Polynomial, 2: Exponential                                                       #
moisture_change_coeff_track_2   0.0008 0.000 0.00       # Moisture Change Model Coefficients: A, B, C (Ax^2+Bx+C, Ae^(Bx)+C) for slave track 2              #
moisture_change_model_track_3   1                       # 0: No change, 1: Polynomial, 2: Exponential                                                       #
moisture_change_coeff_track_3   0.0 0.01 0.0            # Moisture Change Model Coefficients: A, B, C (Ax^2+Bx+C, Ae^(Bx)+C) for slave track 3              #
moisture_change_model_track_4   1                       # 0: No change, 1: Polynomial, 2: Exponential                                                       #
moisture_change_coeff_track_4   0.0 0.0 0.0             # Moisture Change Model Coefficients: A, B, C (Ax^2+Bx+C, Ae^(Bx)+C) for slave track 4              #
```

FIGURE 11. Example input parameter file

- **azimuth_resolution**: This parameter defines the azimuth resolution of the radar (presumed to be half the antenna width) given in meters. The definition of resolution for the purposes of **PolSARproSim**$^+$is width of the point spread function at half height in power
- **slant_range_resolution**: This parameter defines the slant-range resolution of the radar (related to radar system bandwidth) in meters. As with azimuth resolution this is defined as the width of the point spread function at half height in power. The slant-range resolution is held fixed for each track, which corresponds to constant system bandwidth.
- **PSF_broadening_factor**: This parameter defines shape of the system point spread function (see Section 6.6), as a double precision number between 0 and 1. A value of 1.0 leads to a Sinc function while a value of 0.0 leads to a Hanning point spread function.
- **azimuth_sampling_factor**: This double precision parameter defines sampling of the SAR image in the azimuth dimension as a ratio of the pixel dimension to the azimuthal resolution.
- **gnd_range_sampling_factor**: This double precision parameter defines sampling of the SAR image in the ground range dimension as a ratio of the pixel dimension to the ground range resolution.
- **NESZ**: This four dimensional vector defines the noise equivalent sigma zero (in decibels) for the four radar channels HH, HV, VH and VV in that order. The values, separated by spaces or tabs, are used to estimate the amount of thermal noise added to each image.

The following set of parameters are used to define properties of the surface

- **input_dem**: This integer flag (0 or 1) is used to specify whether to use an external DEM. When set (to 1), an external floating point DEM raster file must be supplied. When unset (i.e. 0), a tilted surface is generated internally.
- **external_DEM_file**: This is the path and name of the external DEM file (Unused if input_dem is set to 0). The DEM file must be a binary 4-byte precision floating point raster. The following parameters are used to specify the DEM properties. The center of the DEM is assumed to be at the scene center. **PolSARproSim**$^+$trims and resamples the external DEM to match pixel spacing with the simulated scene.
- **DEM_azimuth_pixels**: The number of pixels of the external DEM file in azimuth. Unused if input_dem is set to 0.
- **DEM_range_pixels**: The number of pixels of the external DEM file in ground range. Unused if input_dem is set to 0.
- **DEM_azimuth_resolution**: The azimuthal resolution (in meters) of the external DEM. This parameter can be changed to change the extent of the DEM used in simulation. Unused if input_dem is set to 0.
- **DEM_range_resolution**: The ground range resolution (in meters) of the external DEM. This parameter can be changed to change the extent of the DEM used in simulation. Unused if input_dem is set to 0.

- **ground_slope_azimuth**: The azimuthal slope (rise over run) of the internally generated sloped surface. This parameter is unused if input_dem is set to 1.
- **ground_slope_range**: The ground range slope (rise over run) of the internally generated sloped surface. This parameter is unused if input_dem is set to 1.
- **ground_moisture**: This is an n-dimensional vector of double precision (for n-tracks) with each entry specifying the amount of moisture in the respective track, with 0 the driest and 10 the wettest soil. The default value is 0 (used if any value is not specified).
- **roughness_model**: An int parameter used to define the ground surface properties. This parameter has the minimum value zero (0), corresponding to the smoothest surface, and the maximum value ten (10), for the roughest surface. This value is translated into actual values of surface height standard deviation and correlation length within **PolSARproSim**$^+$.
- **input_forest**: This integer flag (0 or 1) is used to specify whether to use an external forest. When set (to 1), an external forest XML definition file must be supplied. When unset (i.e. 0), a random forest is generated internally.
- **forest_file**: This is the name (including path) of the XML file that defines the forest using some limited primitives. A more detailed description of this file is given later. Unused if input_forest is set to 0.
- **allometry_file**: This is the name (including path) of the XML file that defines species properties. A more detailed description of this file is given later. This file is mandatory.
- **global_tree_species**: The ascii code specified here is used to look up the species of trees in the internally generated forest from the allometry file when the input_forest flag is set to 0. The species code must be present in the allometry file. Unused if input_forest is set to 1.
- **global_tree_height**: The double precision mean height of trees in the internally generated forest. Unused if input_forest is set to 1.
- **forest_stand_area**: The desired area of the forest stand in square meters. Unused if input_forest is set to 1.
- **stem_density**: The desired stem density of the forest stand in number of stems per hectare. Unused if input_forest is set to 1.
- **forest_output**: This is the name (including path) of the output XML file of the internally generated forest. This file will have the same format as the input forest XML file. Unused if input_forest is set to 1, no output XML file is generated in that case either.
- **draw_forest**: This integer flag (0 or 1) is used to specify whether to create a bitmap image of the forest (internal or external). When set (to 1), two files, forest_image.bmp and forest_image.gri are created, each with a visualization of the forest from the perspective of the radar. NOTE: This option is slow and buggy for large forests so it is recommended it only be enabled for a few trees.

The following set of parameters are used to simulate temporal decorrelation. Since temporal decorrelation can be caused by either movement of scatterers or by moisture

change, the parameters in this group use a nomenclature with ÔchangeÕ as the prefix to accommodate both moisture and motion change.

- **change_height_reference**: This integer flag (0 or 1) is used for choosing the reference height of the change polynomial, i.e. ground or canopy bottom. When set to 0, scatterer height (the independent variable of the motion or moisture profile) is measured from the ground, whereas when it is set to 1, the scatterer height is measured relative to the bottom of the tree crown.
- **position_change_model_track_n**: This integer is used to determine which model will be used to calculate the motion of branches and leaves as a function of height. When set to 0 branch and leaf positions are not changed, when set to 1 a second order polynomial as a function of height (relative to ground or canopy bottom) is used to model scatterer motion, while an exponential function is chosen when this parameter is set to 2. To differentiate tracks, the number of each track is appended to the keyword position_change_model_track, for instance position_change_model_track_1 refers to the model type for the first slave track and so on.
- **position_change_model_coeff_n**: This 3-dimensional vector of double precision is used to specify coefficients of the position change model. For a polynomial model, these are parameters A, B and C for the standard deviation given by: $\sigma = Ah^2 + Bh + C$ and $\sigma = Ae^{Bh} + C$ an exponential model. To differentiate between tracks, the number of each track is appended to the keyword position_change_model_coeff, for instance position_change_model_coeff_1 refers to the model coefficients for the first slave track and so on.
- **moisture_change_model_track_n**: This integer is used to determine which model will be used to calculate the mean moisture offset of branches and leaves as a function of height. When set to 0 branch and leaf moisture values are not changed, when set to 1 a second order polynomial as a function of height (relative to ground or canopy bottom) is used to model moisture offsets, while an exponential function is chosen when this parameter is set to 2. To differentiate between tracks, the number of each track is appended to the keyword moisture_change_model_track, for instance moisture_change_model_track_1 refers to the model type for the first slave track and so on.
- **moisture_change_model_coeff_n**: This 3-dimensional vector of double precision is used to specify coefficients of the moisture offset model. For a polynomial model, these are parameters A, B and C for the mean offset given by: $M_g = Ah^2 + Bh + C$ and $M_g = Ae^{Bh} + C$ an exponential model. To differentiate between tracks, the number of each track is appended to the keyword moisture_change_model_coeff, for instance moisture_change_model_coeff_1 refers to the model coefficients for the first slave track and so on.

### 7.3. **Forest Definition File.**

It is possible to specify an external forest in **PolSARproSim**$^+$using an XML database file.

```
<?xml version="1.0" encoding="ISO-8859-15"?>
<?xml-stylesheet type="text/xsl" href="forest.xsl"?>
<forest>
        <meta>
                <num_trees      units="none">       5           </num_trees>            <!-- The total number of trees in this file                    -->
                <biome          units="none">   temperate       </biome>                <!-- The biome this forest belongs to                          -->
                <area           units="ha">        1.0000       </area>                 <!-- The total area of the forest                              -->
        </meta>
        <tree>
                <species        units="none">        QURU       </species>              <!-- (see allometry file)                                      -->
                <x_location     units="meters">    -12.1400     </x_location>           <!-- X-axis location of tree, scene center is 0.00             -->
                <y_location     units="meters">     1.0000      </y_location>           <!-- Y-axis location of tree, scene center is 0.00             -->
                <tree_height    units="meters">    25.0000      </tree_height>          <!-- Vertical extent from tree base to canopy top              -->
                <canopy_radius  units="meters">     8.2990      </canopy_radius>        <!-- The largest horizontal radius of the canopy               -->
                <canopy_depth   units="meters">    16.8240      </canopy_depth>         <!-- The depth of the live part of the canopy                  -->
                <dry_depth      units="meters">     4.8320      </dry_depth>            <!-- The depth of the dry part of the canopy                   -->
                <stem_diameter  units="meters">     0.5734      </stem_diameter>        <!-- Diameter of trunk at breast height (dbh)                  -->
        </tree>
        <tree>
                <species        units="none">        RHMA       </species>              <!-- (see allometry file)                                      -->
                <x_location     units="meters">     -5.8400     </x_location>           <!-- X-axis location of tree, scene center is 0.00             -->
                <y_location     units="meters">     1.0000      </y_location>           <!-- Y-axis location of tree, scene center is 0.00             -->
                <tree_height    units="meters">    14.2710      </tree_height>          <!-- Vertical extent from tree base to canopy top              -->
                <canopy_radius  units="meters">     3.2500      </canopy_radius>        <!-- The largest horizontal radius of the canopy               -->
                <canopy_depth   units="meters">     9.6140      </canopy_depth>         <!-- The depth of the live part of the canopy                  -->
                <dry_depth      units="meters">     3.0877      </dry_depth>            <!-- The depth of the dry part of the canopy                   -->
                <stem_diameter  units="meters">     0.2352      </stem_diameter>        <!-- Diameter of trunk at breast height (dbh)                  -->
        </tree>
        <tree>
                <species        units="none">        RHMA       </species>              <!-- (see allometry file)                                      -->
                <x_location     units="meters">     0.8400      </x_location>           <!-- X-axis location of tree, scene center is 0.00             -->
                <y_location     units="meters">     1.0000      </y_location>           <!-- Y-axis location of tree, scene center is 0.00             -->
                <tree_height    units="meters">    18.0900      </tree_height>          <!-- Vertical extent from tree base to canopy top              -->
                <canopy_radius  units="meters">     3.8770      </canopy_radius>        <!-- The largest horizontal radius of the canopy               -->
                <canopy_depth   units="meters">    12.0990      </canopy_depth>         <!-- The depth of the live part of the canopy                  -->
                <dry_depth      units="meters">     5.0141      </dry_depth>            <!-- The depth of the dry part of the canopy                   -->
                <stem_diameter  units="meters">     0.2243      </stem_diameter>        <!-- Diameter of trunk at breast height (dbh)                  -->
        </tree>
        <tree>
                <species        units="none">        RHMA       </species>              <!-- (see allometry file)                                      -->
                <x_location     units="meters">     5.5400      </x_location>           <!-- X-axis location of tree, scene center is 0.00             -->
                <y_location     units="meters">     1.5000      </y_location>           <!-- Y-axis location of tree, scene center is 0.00             -->
                <tree_height    units="meters">    10.8080      </tree_height>          <!-- Vertical extent from tree base to canopy top              -->
                <canopy_radius  units="meters">     2.7560      </canopy_radius>        <!-- The largest horizontal radius of the canopy               -->
                <canopy_depth   units="meters">    5.0830       </canopy_depth>         <!-- The depth of the live part of the canopy                  -->
                <dry_depth      units="meters">     1.7261      </dry_depth>            <!-- The depth of the dry part of the canopy                   -->
                <stem_diameter  units="meters">     0.3703      </stem_diameter>        <!-- Diameter of trunk at breast height (dbh)                  -->
        </tree>
        <tree>
                <species        units="none">        PIRE       </species>              <!-- (see allometry file)                                      -->
                <x_location     units="meters">    10.0400      </x_location>           <!-- X-axis location of tree, scene center is 0.00             -->
                <y_location     units="meters">     1.0000      </y_location>           <!-- Y-axis location of tree, scene center is 0.00             -->
                <tree_height    units="meters">    27.3380      </tree_height>          <!-- Vertical extent from tree base to canopy top              -->
                <canopy_radius  units="meters">     3.4670      </canopy_radius>        <!-- The largest horizontal radius of the canopy               -->
                <canopy_depth   units="meters">    19.1380      </canopy_depth>         <!-- The depth of the live part of the canopy                  -->
                <dry_depth      units="meters">     1.3054      </dry_depth>            <!-- The depth of the dry part of the canopy                   -->
                <stem_diameter  units="meters">     0.5813      </stem_diameter>        <!-- Diameter of trunk at breast height (dbh)                  -->
        </tree>
</forest>
```

FIGURE 12. Example Forest Definition File

Figure 12 shows an example forest definition file. As in a typical XML file, all attributes are defined between *tags* that are nested under the root element, which for this file is '`<forest>`'. Two different types of elements are nested within the root element; '`<meta>`' and '`<tree>`'. The first, '`<meta>`' tag contains general information applicable to the entire forest, while the latter tag, '`<tree>`', contains the primitives for each tree. Within the '`<meta>`' tag, the most important element, '`<num_trees>`' is used to define the number of trees to be used in the simulation. A forest definition file can have many more trees than the number specified here, but only as many trees as specified here will be simulated. In Figure 12 for instance this number is 5, so **PolSARproSim**$^+$will simulate a forest with the first five trees specified by the '`<tree>`' tags. If the number had been 3 instead, a forest with only the first three trees would be simulated. If, however, the number specified here incorrectly exceeds the total number of '`<tree>`' tags contained in this file, **PolSARproSim**$^+$defaults to the maximum number of '`<tree>`' tags it encounters. So for instance, if the '`<num_trees>`' in the file in Figure 12 had been set to 12, **PolSARproSim**$^+$would only still simulate a forest with five trees because there are only five '`<tree>`' tags in this particular

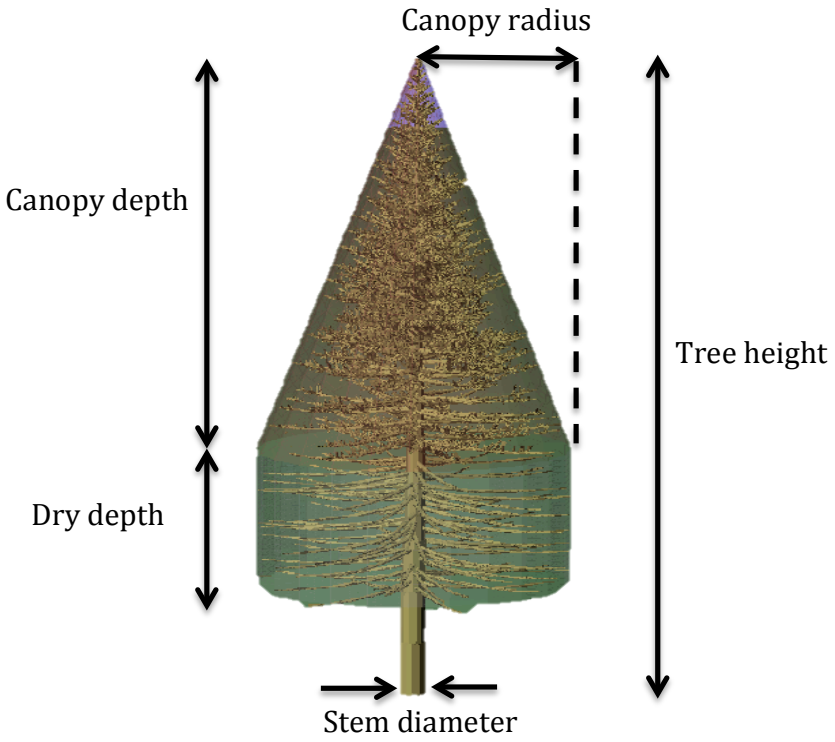


FIGURE 13. Primitives used to define a tree

example. Each tree is defined by eight primitives; species, $x-$ and $y-$ location, height, canopy radius, depth of the live canopy, depth of the dry canopy and stem diameter (or dbh) specified by their respective tags contained within a '`<tree>`' tag. Figure 13 shows the structural variables used to define the structure of a particular tree. Each one of the tags used to define tree attributes is mandatory, **PolSARproSim**$^+$exits with an error if any one parameter is not present. The species of the tree, defined by a string valued element within the '`<species>`' tag, is used to look up its allometric characteristics defined in the allometry XML file (discussed in Section 7.4). If the value specified here for any tree isn't found in the allometry file, **PolSARproSim**$^+$exits with an error. Each tree is explicitly located in the scene through an $x-$ and a $y-$ location specified in the '`<x_location>`' and '`<y_location>`' tags respectively. These two double precision numbers are the azimuth and ground range offsets (in meters) from the scene center, assumed to be a $x = 0$, $y = 0$. The tree height, defined from the base of the tree to the top of the canopy in meters is specified as a double precision number within the tag '`<tree_height>`'. **PolSARproSim**$^+$defines two different types of crowns for a tree in order to differentiate between a live and a dry crown (where branches might be dead and thus have different electrical properties). The canopy radius in meters, specified as a double precision number within the tag '`<canopy_radius>`' applies to both the live and dead crowns, while the depth of each crown can be specified separately within the tags '`<canopy_depth>`' for the live crown and '`<dry_depth>`' for the dry crown as

double precision numbers in meters. Finally, the diameter of the tree trunk at its base (in meters) is input using the '`<stem_diameter>`' tag.

### 7.4. Species Allometry File.

This XML database file is used to define species attributes. A tree in **PolSARproSim**$^+$, is a complex object defined by a whole slew of allometric parameters. They control the fine structural details of a particular realization of a tree such as the number, shape and electrical properties of crowns, stems branches and leaves. This allows user the flexibility to define any number of tree species in significant detail. Since the number of parameters needed to define a particular species is quite high, the structure of the species allometry file is designed so the user can define a species without having to define each and every parameter. This is accomplished by letting each species belong to a general category, say for instance deciduous or coniferous. While all allometric parameters must be defined for a category, parameters for a particular species are adopted from its category unless explicitly specified in the species tag.

The allometry file consists of four main types of tags nested inside the root element '`<allometry>`'. Two of them, titled '`<species_list>`' and '`<category_list>`' contain the names of species and categories defined in this file respectively, while the other two contain all the allometric parameters for each species or category. Figure 14 shows an example of the category and species lists. Each entry in either the species or

```
<species_list>                                                          <!-- MANDATORY list of species                  -->
    <name              units="none">   HEDGE       </name>              <!-- 0: The hedge                               -->
    <name              units="none">   TSCA        </name>              <!-- 1: Tsuga canadensis, Eastern Hemlock       -->
    <name              units="none">   PIRE        </name>              <!-- 2: Pinus resinosa, Red Pine                -->
    <name              units="none">   BEPA        </name>              <!-- 3: Betula papyrifera, Paper/White Birch    -->
    <name              units="none">   ACRU        </name>              <!-- 4: Acer rubrum, Red Maple                  -->
    <name              units="none">   ACSA        </name>              <!-- 5: Acer saccarum, Sugar Maple              -->
    <name              units="none">   ACPE        </name>              <!-- 6: Acer pennsylvanicum, Striped Maple      -->
    <name              units="none">   BEAL        </name>              <!-- 7: Betula alleghensis,  Yellow Birch       -->
    <name              units="none">   BELE        </name>              <!-- 8: Betula lenta, Black Birch               -->
    <name              units="none">   BEPO        </name>              <!-- 9: Betula populifolia, Grey Birch          -->
    <name              units="none">   FAGR        </name>              <!-- 10: Fagus grandifolia, American Beach      -->
    <name              units="none">   OSVI        </name>              <!-- 11: Ostrya virginiana, Hophornbeam         -->
    <name              units="none">   PIRU        </name>              <!-- 12: Picea rubens, Red Spruce               -->
    <name              units="none">   PIST        </name>              <!-- 13: Pinus strobus, White Pine              -->
    <name              units="none">   PRSE        </name>              <!-- 14: Prunus serotina, Black Cherry          -->
    <name              units="none">   QUAL        </name>              <!-- 15: Quercus alba, White Oak                -->
    <name              units="none">   QURU        </name>              <!-- 16: Quercus rubra, Red Oak                 -->
    <name              units="none">   SNAG        </name>              <!-- 17: Unknown snaggy stuff                   -->
    <name              units="none">   RHMA        </name>              <!-- 17: Rhizophora mangle, Red Mangrove        -->
</species_list>
<category_list>                                                         <!-- MANDATORY list of categories               -->
    <name              units="none">   deciduous   </name>              <!-- Broadleaf, deciduous trees                 -->
    <name              units="none">   coniferous  </name>              <!-- Needled, evergreen coniferous trees        -->
</category_list>
```

FIGURE 14. Example of species and category tags in the species allometry XML file

category list is an ASCII string within the '`<name>`' tag, for example the 'HEDGE' or the 'TSCA' species or the 'deciduous' or 'coniferous' categories. Figure 15 shows all the parameters needed to define a tree in **PolSARproSim**$^+$for the 'deciduous' category from the example shown in Figure 14. All parameters must be nested within a tag named for the particular category, which in this case is '`<deciduous>`'. For the example here a similar tag with every parameter must also exist for the coniferous category as defined in Figure 14. (Note: XML tags are case sensitive so the name in the category or species list must match their respective tags exactly). Finally, each species named in the species list must also have a corresponding tag, e.g. '`<HEDGE>`', '`<TSCA>`',

```
<deciduous>
        <crown_live_shape              units="none">   spheroid        </crown_live_shape>                  <!-- All fields MANDATORY if category defined in category_list  -->
        <crown_dry_shape               units="none">   none            </crown_dry_shape>                   <!-- shape of living crown (conical/spheroid/cylindrical)       -->
        <crown_live_depth_factor       units="none">   0.5             </crown_live_depth_factor>           <!-- shape of dry crown (concial/spheroid/cylindrical/none)     -->
        <crown_dry_depth_factor        units="none">   0.5             </crown_dry_depth_factor>            <!-- fraction of live crown depth at which crown radius is max. -->
        <crown_dry_offset_factor       units="none">   0               </crown_dry_offset_factor>           <!-- fraction of dry crown depth at which crown radius is max.  -->
        <crown_live_alpha              units="none">   0.410           </crown_live_alpha>                  <!-- offset dry crown from live by a fraction of the bare stem. -->
        <crown_dry_alpha               units="none">   0.000           </crown_dry_alpha>                   <!-- to est. live crown depth (unless forest primitives used)   -->
        <crown_radius_factor           units="none">   0.410           </crown_radius_factor>               <!-- to est. dry crown depth (unless forest primitives used)    -->
        <stem_start_radius_factor      units="none">   0.7925          </stem_start_radius_factor>          <!-- to est. crown radius (unless forest primitives used)       -->
        <stem_maximum_polar_angle      units="deg">    10.0            </stem_maximum_polar_angle>          <!-- to est. stem start radius (unless forest primitives used)  -->
        <stem_radius_end_factor        units="none">   0.7345          </stem_radius_end_factor>            <!-- maximum stem angle (from vertical)                         -->
        <stem_tropism_factor           units="none">   0.2             </stem_tropism_factor>               <!-- factor to determine stem end radius as function of height  -->
        <stem_lamdacx_stdev            units="none">   0.25            </stem_lamdacx_stdev>                <!-- factor to determine stem tropism or curvature              -->
        <stem_lamdacx_mean             units="none">   0.75            </stem_lamdacx_mean>                 <!-- std dev of stem curvature wavelength function in x-dim.    -->
        <stem_lamdacy_stdev            units="none">   0.25            </stem_lamdacy_stdev>                <!-- mean of stem curvature wavelength function in x-dim.       -->
        <stem_lamdacy_mean             units="none">   0.75            </stem_lamdacy_mean>                 <!-- std dev of stem curvature wavelength function in y-dim.    -->
        <stem_gamma_stdev              units="none">   0.015           </stem_gamma_stdev>                  <!-- mean of stem curvature wavelength function in y-dim.       -->
        <stem_gamma_mean               units="none">   0.045           </stem_gamma_mean>                   <!-- std dev of stem curvature scale-factor function            -->
        <stem_moisture                 units="none">   0.370           </stem_moisture>                     <!-- mean of stem curvature scale-factor function               -->
        <stem_tropism_polar_angle      units="deg">    0.0             </stem_tropism_polar_angle>          <!-- gravimetric moisture content of stem 0<Mg<0.7              -->
        <stem_tropism_azimuth_angle    units="deg">    0.0             </stem_tropism_azimuth_angle>        <!-- Polar angle (theta) of the stem tropism direction          -->
        <primary_polar_angle_mean      units="deg">    50.0            </primary_polar_angle_mean>          <!-- Azimuthal angle (phi) of the stem tropism direction        -->
        <primary_polar_angle_stdev     units="deg">    25.0            </primary_polar_angle_stdev>         <!-- Average primary branch polar angle (from vertical)         -->
        <primary_layer_density         units="none">   2.01            </primary_layer_density>             <!-- Std. dev. of primary branch polar angle (from vertical)    -->
        <primary_sections              units="none">   2               </primary_sections>                  <!-- Primary layer density per unit depth (of canopy)           -->
        <primary_azimuth_factor        units="deg">    90              </primary_azimuth_factor>            <!-- Average no. of sections per layer of primary branches      -->
        <primary_radius_quadratic_coeff units="none">  0.0             </primary_radius_quadratic_coeff>    <!-- Azimuth angle std. factor for primary branches             -->
        <primary_radius_linear_coeff   units="none">   -0.3465         </primary_radius_linear_coeff>       <!-- Quadratic parameter for the primary start radius function  -->
        <primary_radius_constant_coeff units="none">   0.693           </primary_radius_constant_coeff>     <!-- Linear parameter for the primary start radius function     -->
        <primary_tropism_factor        units="none">   0.3             </primary_tropism_factor>            <!-- Constant parameter for the primary start radius function   -->
        <primary_tropism_polar_angle   units="deg">    0.0             </primary_tropism_polar_angle>       <!-- factor to determine primary tropism                        -->
        <primary_tropism_azimuth_angle units="deg">    0.0             </primary_tropism_azimuth_angle>     <!-- Polar angle (theta) of the primary tropism direction       -->
        <primary_lamdacx_stdev         units="none">   0.25            </primary_lamdacx_stdev>             <!-- Azimuthal angle (phi) of the primary tropism direction     -->
        <primary_lamdacx_mean          units="none">   0.75            </primary_lamdacx_mean>              <!-- std dev of primary curvature wavelength function in x-dim. -->
        <primary_lamdacy_stdev         units="none">   0.25            </primary_lamdacy_stdev>             <!-- mean of primary curvature wavelength function in x-dim.    -->
        <primary_lamdacy_mean          units="none">   0.75            </primary_lamdacy_mean>              <!-- std dev of primary curvature wavelength function in y-dim. -->
        <primary_gamma_stdev           units="none">   0.025           </primary_gamma_stdev>               <!-- mean of primary curvature wavelength function in y-dim.    -->
        <primary_gamma_mean            units="none">   0.075           </primary_gamma_mean>                <!-- std dev of primary curvature scale-factor function         -->
        <primary_live_moisture         units="none">   0.424           </primary_live_moisture>             <!-- mean of primary curvature scale-factor function            -->
        <primary_dry_moisture          units="none">   0.100           </primary_dry_moisture>              <!-- gravimetric moisture of live primary branches 0<Mg<0.7     -->
        <primary_dry_layer_density     units="none">   0.00            </primary_dry_layer_density>         <!-- gravimetric moisture of dry primary branches 0<Mg<0.7      -->
        <primary_dry_sections          units="none">   0               </primary_dry_sections>              <!-- Dry primary layer density per unit depth (of dry canopy)   -->
        <primary_dry_polar_angle_mean  units="deg">    50.0            </primary_dry_polar_angle_mean>      <!-- Number of sections per layer of dry primary branches       -->
        <primary_dry_polar_angle_stdev units="deg">    25.0            </primary_dry_polar_angle_stdev>     <!-- Average dry primary branch polar angle (from vertical)     -->
        <secondary_number_intercept    units="none">   0               </secondary_number_intercept>        <!-- Stdev of dry primary branch polar angle (from vertical)    -->
        <secondary_number_slope        units="none">   1.045           </secondary_number_slope>            <!-- Intercept of function determining secondary branch numbers -->
        <secondary_offset_fraction     units="none">   0.35            </secondary_offset_fraction>         <!-- Slope of function determining secondary branch numbers     -->
        <secondary_polar_angle_mean    units="deg">    45.00           </secondary_polar_angle_mean>        <!-- Fractional offset for the starting location of secondaries -->
        <secondary_polar_angle_stdev   units="deg">    20.00           </secondary_polar_angle_stdev>       <!-- Average secondary branch polar angle (from primary)        -->
        <secondary_radius_factor       units="none">   0.75            </secondary_radius_factor>           <!-- Std. dev. of secondary branch polar angle (from primary)   -->
        <secondary_tropism_factor      units="none">   0.6             </secondary_tropism_factor>          <!-- Secondary start radius as a factor of the primary radius   -->
        <secondary_tropism_polar_angle units="deg">    0.0             </secondary_tropism_polar_angle>     <!-- factor to determine secondary tropism                      -->
        <secondary_tropism_azimuth_angle units="deg">  0.0             </secondary_tropism_azimuth_angle>   <!-- Polar angle (theta) of the secondary tropism direction     -->
        <secondary_lamdacx_stdev       units="none">   0.25            </secondary_lamdacx_stdev>           <!-- Azimuthal angle (phi) of the secondary tropism direction   -->
        <secondary_lamdacx_mean        units="none">   0.75            </secondary_lamdacx_mean>            <!-- std dev of secondary curvature wavelength function x-dim.  -->
        <secondary_lamdacy_stdev       units="none">   0.25            </secondary_lamdacy_stdev>           <!-- mean of secondary curvature wavelength function in x-dim.  -->
        <secondary_lamdacy_mean        units="none">   0.75            </secondary_lamdacy_mean>            <!-- std dev of secondary curvature wavelength function y-dim.  -->
        <secondary_gamma_stdev         units="none">   0.025           </secondary_gamma_stdev>             <!-- mean of secondary curvature wavelength function in y-dim.  -->
        <secondary_gamma_mean          units="none">   0.075           </secondary_gamma_mean>              <!-- std dev of secondary curvature scale-factor function       -->
        <tertiary_minimum_length       units="m">      0.0             </tertiary_minimum_length>           <!-- mean of secondary curvature scale-factor function          -->
        <tertiary_minimum_radius       units="m">      0.0             </tertiary_minimum_radius>           <!-- minimum length of tertiary branches                        -->
        <tertiary_volume_fraction      units="none">   0.001           </tertiary_volume_fraction>          <!-- minimum radius of tertiary branches                        -->
        <tertiary_number_factor        units="none">   120.0           </tertiary_number_factor>            <!-- Fraction of crown volume occupied by tertiary branches     -->
        <tertiary_moisture             units="none">   0.4735          </tertiary_moisture>                 <!-- Factor to scale the number of tertiary branches imaged     -->
        <leaf_shape                    units="none">   broadleaf       </leaf_shape>                        <!-- gravimetric moisture of tertiary branches 0<Mg<0.7         -->
        <leaf_width_mean               units="m">      0.060           </leaf_width_mean>                   <!-- shape of tree leaves (broadleaf/needle)                    -->
        <leaf_width_stdev              units="m">      0.030           </leaf_width_stdev>                  <!-- mean leaf width                                            -->
        <leaf_height_mean              units="m">      0.040           </leaf_height_mean>                  <!-- leaf width standard deviation                              -->
        <leaf_height_stdev             units="m">      0.020           </leaf_height_stdev>                 <!-- mean leaf height                                           -->
        <leaf_thickness_mean           units="m">      0.001           </leaf_thickness_mean>               <!-- leaf height standard deviation                             -->
        <leaf_thickness_stdev          units="m">      0.001           </leaf_thickness_stdev>              <!-- mean leaf thickness                                        -->
        <leaf_volume_fraction          units="none">   0.00025         </leaf_volume_fraction>              <!-- leaf thickness standard deviation                          -->
        <leaf_number_factor            units="none">   30.0            </leaf_number_factor>                <!-- Fraction of crown volume occupied by leaves                -->
        <leaf_moisture                 units="none">   0.5500          </leaf_moisture>                     <!-- Factor to scale the number of leaves imaged                -->
</deciduous>                                                                                                <!-- gravimetric moisture content of leaves 0<Mg<0.7            -->
```

FIGURE 15. Example of allometric parameters for a category tag in the species allometry XML file

'`<PIRE>`' etc., however in this case the only mandatory parameter is the category that the particular species belong to. Figure 16 shows an example of a species (`HEDGE`) tag

```
<HEDGE>
        <category                      units="none">   deciduous      </category>                          <!-- MANDATORY category of trees (unused if all fields defined) -->
        <crown_live_shape              units="none">   cylindrical     </crown_live_shape>                  <!-- shape of living crown (conical/spheroid/cylindrical)       -->
        <crown_live_depth_factor       units="none">   1               </crown_live_depth_factor>           <!-- fraction of live crown depth at which crown radius is max. -->
        <stem_maximum_polar_angle      units="deg">    1.0             </stem_maximum_polar_angle>          <!-- maximum stem angle (from vertical)                         -->
        <primary_dry_layer_density     units="none">   0.00            </primary_dry_layer_density>         <!-- Dry primary layer density per unit depth (of dry canopy)   -->
        <primary_dry_sections          units="none">   0               </primary_dry_sections>              <!-- Number of sections per layer of dry primary branches       -->
        <secondary_number_intercept    units="none">   0.0             </secondary_number_intercept>        <!-- Intercept of function determining secondary branch numbers -->
        <secondary_number_slope        units="none">   0.0             </secondary_number_slope>            <!-- Slope of function determining secondary branch numbers     -->
        <tertiary_minimum_length       units="m">      0.5             </tertiary_minimum_length>           <!-- minimum length of tertiary branches                        -->
        <tertiary_minimum_radius       units="m">      0.011           </tertiary_minimum_radius>           <!-- minimum radius of tertiary branches                        -->
        <tertiary_volume_fraction      units="none">   0.004           </tertiary_volume_fraction>          <!-- Fraction of crown volume occupied by tertiary branches     -->
        <tertiary_number_factor        units="none">   120.0           </tertiary_number_factor>            <!-- Factor to scale the number of tertiary branches imaged     -->
        <leaf_width_mean               units="m">      0.090           </leaf_width_mean>                   <!-- mean leaf width                                            -->
        <leaf_width_stdev              units="m">      0.010           </leaf_width_stdev>                  <!-- leaf width standard deviation                              -->
        <leaf_height_mean              units="m">      0.090           </leaf_height_mean>                  <!-- mean leaf height                                           -->
        <leaf_height_stdev             units="m">      0.040           </leaf_height_stdev>                 <!-- leaf height standard deviation                             -->
        <leaf_thickness_mean           units="m">      0.001           </leaf_thickness_mean>               <!-- mean leaf thickness                                        -->
        <leaf_thickness_stdev          units="m">      0.0005          </leaf_thickness_stdev>              <!-- leaf thickness standard deviation                          -->
        <leaf_volume_fraction          units="none">   0.000125        </leaf_volume_fraction>              <!-- Fraction of crown volume occupied by leaves                -->
        <leaf_number_factor            units="none">   30.0            </leaf_number_factor>                <!-- Factor to scale the number of leaves imaged                -->
        <leaf_moisture                 units="none">   0.5500          </leaf_moisture>                     <!-- gravimetric moisture content of leaves 0<Mg<0.7            -->
</HEDGE>
```

FIGURE 16. Example of allometric parameters for a species tag in the species allometry XML file

defined in the allometry file. Note that the parameters here are only a subset of the

parameters listed in the '<deciduous>' tag shown in Figure 15. The only mandatory parameter is the category (<category>), all other parameters can be omitted. However, since this particular species differs significantly from the general deciduous category, a few parameters need to be explicitly defined which will take precedence over the parameters from the deciduous category. A description of each allometric parameter follows.

7.4.1. *Crown Related Parameters.* This first set of parameters is related to defining the tree crown. In **PolSARproSim**$^+$a crown is a geometrical object that will contain all branches and leaves. The shape of the crown will decide the size and extent of branches that are grown only until they reach the crown. Furthermore they are These are

- <crown_live_shape>: Shape of the living crown. An ASCII string with three possible values conical, spheroid or cylindrical.
- <crown_dry_shape>: Shape of the dry crown. An ASCII string with four possible values conical, spheroid, cylindrical or none. Unlike the live crown that is necessary for every tree, the dry crown can be removed by setting this parameter to 'none'.
- <crown_live_depth_factor>: Fraction of the crown depth at which the crown radius reaches the specified value. Double precision unit-less number between 0.5 and 1. For spherical crown, if this factor is 0.5, the crown radius will be maximum at half the crown depth producing a spherical crown. On the other extreme the crown radius is maximum at the bottom of the crown for a value of 1 for this factor, producing a hemispherical crown. This factor has no effect on cylindrical crowns, and should be set to 1.0 for conical crowns.
- <crown_dry_depth_factor>: Same as <crown_live_depth_factor> except for the dry crown.
- <crown_dry_offset_factor>: This parameter, a double precision unit-less number between 0 and 1 is used to offset the dry crown from the living crown. At a value of 0.0, the dry crown abuts the living crown, while at a value of 1.0 the dry crown sits on the ground separated from the living crown by the bare stem.
- <crown_live_alpha>: When **PolSARproSim**$^+$generates a forest internally (i.e. when input_forest in the input parameter file is set to 0), this parameter, $\alpha$, is used to estimate the canopy depth (in meters) as a function of the tree height, such that $c_{depth} = \alpha \times h_{tree}$ where $c_{depth}$ is the depth of the living canopy and $h_{tree}$ the mean tree height (in meters).
- <crown_dry_alpha>: Same as <crown_live_alpha> except for the dry crown. Is unused if <crown_dry_shape> is set to 'none'.
- <crown_radius_factor>: When **PolSARproSim**$^+$generates a forest internally (i.e. when input_forest in the input parameter file is set to 0), this parameter, $c_{rf}$, is used to estimate crown radius (in meters) from mean tree height, such that $c_{radius} = c_{rf} \times h_{tree}$, where $c_{radius}$ is the maximum crown radius and $h_{tree}$ the mean tree height (in meters).

7.4.2. *Stem Related Parameters.* The second set of parameters are used to define the stem structure of a species. A stem is assumed to be a single branch, modeled using the equations described in Section 5.2.3. While the tree specific primitives such as location and height etc are either defined in the forest input file, the parameters required to realize a stem are given here

- `<stem_start_radius_factor>`: When **PolSARproSim**$^+$generates a forest internally (i.e. when input_forest in the input parameter file is set to 0), this parameter, $s_{srf}$, is used to estimate stem start radius (in meters) from mean tree height, such that $s_{sr} = s_{srf} \times h_{tree}/100$, where $s_{sr}$ is the stem radius at the base of the tree and $h_{tree}$ the mean tree height (in meters).
- `<stem_radius_end_factor>`: Used to estimate the end radius of the stem as a function of height (both in meters). The end radius, $s_{er}$, is calculated as $s_{er} = s_{erf} \times h_{tree}/1000$, where $s_{erf}$ is the end radius factor and $h_{tree}$ is the tree height.
- `<stem_maximum_polar_angle>`: This is the maximum stem angle from vertical (in degrees), $\theta_{smax}$. The stem direction, $\hat{s}_{z_0}$, is realized as a three dimensional polar vector ($\hat{s}_{z_0} = 1.0\hat{r} + \theta_s\hat{\theta} + \phi_s\hat{\phi}$) using this parameter, such that

$$\theta_s = \cos^{-1}\left(\theta_{srad} + \tilde{x}_1\left(1 - \cos\theta_{srad}\right)\right)$$
$$\phi_s = 2\pi\tilde{x}_2$$

  where $\theta_{srad} = \pi\theta_{smax}/180$ and $\tilde{x}_1, \tilde{x}_2 \sim U\left[0,1\right]$
- `Stem Tropism`: Stem tropism mimics the natural tendency of stems to grow towards light or sag under the effects of gravity. Three parameters define the tropism of a stem, the stem tropism factor `<stem_tropism_factor>`, $d_p$ in (50) and its direction defined by the two polar angles `<stem_tropism_polar_angle>`, $st_\theta$ and the `<stem_tropism_azimuth_angle>`, $st_\phi$ (both in degrees). The tropism factor is a double precision number between 0 and 1, while the tropism direction, $\hat{p}$ in (50) is calculated as

$$\hat{p} = \sin\theta_t\cos\phi_t\hat{x} + \sin\theta_t\sin\phi_t\hat{y} + \cos\phi_t\hat{z}$$

  where $\theta_t = st_\theta \times \pi/180$ and $\phi_t = st_\phi \times \pi/180$
- `Stem Crookedness`: The amount of crookedness of a branch is modeled by a range of parameters as shown in (52), however three of them are in control of the user and specified for the stem using six tags (two for each parameter)
  - `<stem_lambdacx_mean>` ($\overline{\lambda}_{cx}$), `<stem_lambdacx_stdev>` ($\sigma_{\lambda_{cx}}$)
  - `<stem_lambdacy_mean>` ($\overline{\lambda}_{cy}$), `<stem_lambdacy_stdev>` ($\sigma_{\lambda_{cy}}$)
  - `<stem_gamma_mean>` ($\overline{\gamma}$) and `<stem_gamma_stdev>` ($\sigma_\gamma$)

  so that the parameters $\lambda_{cx}$, $\lambda_{cy}$ and $\gamma$ in (52) are realized such that

$$\lambda_{cx} = \overline{\lambda}_{cx} + \tilde{x}_1\sigma_{\lambda_{cx}}$$
$$\lambda_{cy} = \overline{\lambda}_{cy} + \tilde{x}_2\sigma_{\lambda_{cy}}$$
$$\gamma = \overline{\gamma} + \tilde{x}_3\sigma_\gamma$$

  where $\tilde{x}_1,\ \tilde{x}_2,\ \tilde{x}_2 \sim U\left[0,1\right]$.

- `<stem_moisture>`: This double precision parameter between 0 and 0.7 is the gravimetric moisture content ($M_g$) of the stem used to derive its electrical properties, i.e. its effective permittivity, which is estimated using (58) as given in [13].

7.4.3. *Primary Branch Parameters.* Primary branches are assumed to originate at the stem and terminate at the edge of the crown. As an object inside the software, a primary branch is the same structure as a stem, however the rules and parameters for realizing primary branches are different. **PolSARproSim**$^+$distinguishes between two types of primary branches, in the dry crown and in the living crown. Most of the parameters that define either the live or dry primary branches (although not always independently) are under control of the user through the allometry file and are listed here

- `Number of Primaries`: A living tree crown is divided into $N_{layers}$ vertical layers, and $N_{sec}$ sections per layer. The number of sections per layer for the living crown is specified directly in the allometry file as an integer within the tag `<primary_sections>`, while the number of layers is estimated using the parameter `<primary_layer_density>` ($\Delta_N$) such that

$$N_{layers} = \Delta_N \times c_{depth}$$

  where $c_{depth}$ is the canopy depth of the living crown in meters.
- `Primary Start Radius`: The start radius of a primary branch is a function of the location on the stem where it originates, more specifically a quadratic function of the parameter $t$ for a stem as seen in (50). The start radius of a primary branch, $p_{sr}$ is then given by

$$p_{sr} = P_A t^2 + P_B t + P_C$$

  where the coefficients $A$, $B$, and $C$ are specified using tags in the allometry file
  - `<primary_radius_quadratic_coeff>`: The coefficient $P_A$
  - `<primary_radius_linear_coeff>`: The coefficient $P_B$
  - `<primary_radius_constant_coeff>`: The coefficient $P_C$
- `Primary Start Direction`: The mean start direction of the living primary branches is specified using three parameters
  - `<primary_polar_angle_mean>`: The mean polar angle, $p_{\bar{\theta}}$ in degrees.
  - `<primary_polar_angle_stdev>`: The standard deviation of the polar angle, $p_{\sigma_\theta}$ in degrees.
  - `<primary_azimuth_factor>`: The standard deviation of the azimuthal angle, $p_{\sigma_\phi}$ in degrees.

  The primary start direction, $\hat{p}_{z_0}$, is realized as a three dimensional polar vector ($\hat{p}_{z_0} = 1.0\hat{r} + \theta_p\hat{\theta} + \phi_p\hat{\phi}$) such that

$$\theta_p\left(t_b\right) = \bar{\theta} + \frac{\bar{t} - t_b}{\bar{t} - t_{min}}\sigma_\theta^r$$

where $\bar{\theta} = p_{\bar{\theta}} \times \pi/180$, and $\sigma_\theta^r = p_{\sigma_\theta} \times \pi/180$. The parameter $t_b$ has a mean value $\bar{t} = 0.5\,(t_{max} - t_{min})$ with $t_{max}$ and $t_{min}$ the value at the top and bottom of the canopy respectively, and is computed as $t_b = t_{min} + i_{layer}\Delta_{layer} - \tilde{x}\Delta_{layer}$, where $i_{layer}$ is the layer at which the branch started, while $\Delta_{layer} = (t_{max} - t_{min})\,/N_{layers}$ and $\tilde{x} \sim U\,[0,1]$. The azimuthal angle, $\phi_p$ is given by

$$\phi_p = p_{\sigma_\phi} \times \pi/180\,(\tilde{x} - 0.5)$$

where $p_{\sigma_\phi}$ is the primary azimuth factor in degrees and $\tilde{x} \sim U\,[0,1]$.

- `Primary Branch Tropism`: Primary tropism, like stem tropism mimics the natural tendency of branches to grow towards light or sag under the effects of gravity. Three parameters define the tropism of a primary branch, the tropism factor, and its polar angles
  - `<primary_tropism_factor>` The tropism factor is a unit less double precision number between 0 and 1, $d_p$ in (50).
  - `<primary_tropism_polar_angle>` the double precision polar angle, $pt_\theta$, for the tropism direction, in degrees.
  - `<primary_tropism_azimuth_angle>` the double precision azimuthal angle, $pt_\phi$ for the tropism direction, in degrees.

  The tropism direction, $\hat{p}$ in (50) is calculated as

$$\hat{p} = \sin\theta_t \cos\phi_t \hat{x} + \sin\theta_t \sin\phi_t \hat{y} + \cos\phi_t \hat{z}$$

  where $\theta_t = pt_\theta \times \pi/180$ and $\phi_t = pt_\phi \times \pi/180$
- `Primary Branch Crookedness`: The amount of crookedness of a primary branch is modeled by a range of parameters as shown in (52), however three of them are in control of the user and specified for the primary branch using six tags (two for each parameter)
  - `<primary_lambdacx_mean>` ($\overline{\lambda}_{cx}$), `<primary_lambdacx_stdev>` ($\sigma_{\lambda_{cx}}$)
  - `<primary_lambdacy_mean>` ($\overline{\lambda}_{cy}$), `<primary_lambdacy_stdev>` ($\sigma_{\lambda_{cy}}$)
  - `<primary_gamma_mean>` ($\overline{\gamma}$) and `<primary_gamma_stdev>` ($\sigma_\gamma$)

  so that the parameters $\lambda_{cx}$, $\lambda_{cy}$ and $\gamma$ in (52) are realized such that

$$\lambda_{cx} = \overline{\lambda}_{cx} + \tilde{x}_1 \sigma_{\lambda_{cx}}$$
$$\lambda_{cy} = \overline{\lambda}_{cy} + \tilde{x}_2 \sigma_{\lambda_{cy}}$$
$$\gamma = \overline{\gamma} + \tilde{x}_3 \sigma_\gamma$$

  where $\tilde{x}_1,\ \tilde{x}_2,\ \tilde{x}_2 \sim U\,[0,1]$.
- `<primary_live_moisture>`: This double precision parameter between 0 and 0.7 is the gravimetric moisture content ($M_g$) of the primary branches used to derive their electrical properties, i.e. their effective permittivities, which are estimated using (58) as given in [13].
- `<primary_dry_moisture>`: This double precision parameter between 0 and 0.7 is the gravimetric moisture content ($M_g$) of the primary branches in the dry canopy used to derive their electrical properties, i.e. their effective permittivities, which are estimated using (58) as given in [13].

- `Number of Dry Primaries`: The dry crown is divided into $N_{layers}$ vertical layers, and $N_{sec}$ sections per layer. The number of sections per layer for the living crown is specified directly in the allometry file as an integer within the tag `<primary_dry_sections>`, while the number of layers is estimated using the parameter `<primary_dry_layer_density>` ($\Delta_N$) such that

$$N_{layers} = \Delta_N \times c_{depth}$$

  where $c_{depth}$ is the canopy depth of the dry crown in meters.
- `Start Direction of Dry Primaries`: The mean start direction of the dry primary branches is specified using two parameters
  - `<primary_dry_polar_angle_mean>`: The mean polar angle, $p_{\bar{\theta}}$ in degrees.
  - `<primary_dry_polar_angle_stdev>`: The standard deviation of the polar angle, $p_{\sigma_\theta}$ in degrees.

  The primary start direction, $\hat{p}_{z_0}$, is realized as a three dimensional polar vector ($\hat{p}_{z_0} = 1.0\hat{r} + \theta_p\hat{\theta} + \phi_p\hat{\phi}$) such that

$$\theta_p\left(t_b\right) = \bar{\theta} + \frac{\bar{t} - t_b}{\bar{t} - t_{min}}\sigma_\theta^r$$

  where $\bar{\theta} = p_{\bar{\theta}} \times \pi/180$, and $\sigma_\theta^r = p_{\sigma_\theta} \times \pi/180$. The parameter $t_b$ has a mean value $\bar{t} = 0.5\left(t_{max} - t_{min}\right)$ with $t_{max}$ and $t_{min}$ the value at the top and bottom of the dry canopy respectively, and is computed as $t_b = t_{min} + i_{layer}\Delta_{layer} - \tilde{x}\Delta_{layer}$, where $i_{layer}$ is the layer at which the branch started, while $\Delta_{layer} = \left(t_{max} - t_{min}\right)/N_{layers}$ and $\tilde{x} \sim U\left[0, 1\right]$.

7.4.4. *Secondary Branch Parameters.* Secondary branches are assumed to stem from primary branches. Each secondary branch, much like primary branches or stems is defined as a structure using (50) however the rules and parameters for realizing secondary branches are different. While a dry crown is assumed to not have any secondary branches, the following parameters are used to define secondary branches through the allometry input file

- `Number of Secondaries`: The number of secondary branches per primary branch is controlled as a linear function of the length of the primary branch. Two parameters, the slope and intercept of the linear function are specified in the allometry file inside the tags
  - `<secondary_number_intercept>`: The intercept of the linear function, $s_C$, as a double precision unit less number.
  - `<secondary_number_slope>`: The slope of the linear function, $s_M$, as a double precision number in unites of $m^{-1}$.

  The number of secondaries that originate from a particular primary branch is then calculated as

$$s_N = s_M p_{len} + s_C$$

  where $p_{len}$ is the length of the originating primary branch.
- `Secondary Start Direction`: The mean secondary start direction is defined by two parameters, the mean and standard deviation of the polar angle inside the tags

- `<secondary_polar_angle_mean>`: The mean polar angle, $s_{\bar{\theta}}$ in degrees.
- `<secondary_polar_angle_stdev>`: The standard deviation of the polar angle, $s_{\sigma_\theta}$ in degrees.

The secondary start direction, $\hat{s}_{z_0}$, is realized as a three dimensional polar vector ($\hat{s}_{z_0} = 1.0\hat{r} + \theta_s\hat{\theta} + \phi_s\hat{\phi}$) such that

$$\theta_s \sim N\left(\bar{\theta}, \sigma^2_{\theta_r}\right)$$

where $\bar{\theta} = s_{\bar{\theta}} \times \pi/180$, and $\sigma^2_{\theta_r} = \left(s_{\sigma_\theta} \times \pi/180\right)^2$, while $\phi_s = 2\pi\left(\tilde{x} - 0.5\right)$ where $\tilde{x} \sim U\left[0, 1\right]$.

- `Secondary Start Radius`: The start radius of a secondary branch is a function of the location on the primary branch where it originates, more specifically a linear function of the parameter $t$ for a primary branch as seen in (50). The start radius of a secondary branch, $s_{sr}$ is then given by

$$s_{sr} = s_{rf} \times p_r\left(t = t_s\right)$$

where $p_r\left(t = t_s\right)$ is the radius of the primary branch at the location where the secondary branch originates (at $t = t_s$), and the coefficient $s_{rf}$ is specified using `<secondary_radius_factor>` tag in the allometry file as a double precision unit less number between 0 and 1.

- `Secondary Branch Tropism`: Secondary tropism, like stem or primary branch tropism mimics the natural tendency of branches to grow towards light or sag under the effects of gravity. Three parameters define the tropism of a secondary branch, the tropism factor, and its polar angles
  - `<secondary_tropism_factor>` The tropism factor is a unit less double precision number between 0 and 1, $d_s$ in (50).
  - `<secondary_tropism_polar_angle>` the double precision polar angle, $st_\theta$, for the tropism direction, in degrees.
  - `<secondary_tropism_azimuth_angle>` the double precision azimuthal angle, $st_\phi$ for the tropism direction, in degrees.

  The tropism direction, $\hat{p}$ in (50) is calculated as

$$\hat{p} = \sin\theta_t \cos\phi_t \hat{x} + \sin\theta_t \sin\phi_t \hat{y} + \cos\phi_t \hat{z}$$

  where $\theta_t = st_\theta \times \pi/180$ and $\phi_t = st_\phi \times \pi/180$

- `Secondary Branch Crookedness`: The amount of crookedness of a secondary branch is modeled by a range of parameters as shown in (52), however three of them are in control of the user and specified for the secondary branch using six tags (two for each parameter)
  - `<secondary_lambdacx_mean>` ($\overline{\lambda}_{cx}$), `<secondary_lambdacx_stdev>` ($\sigma_{\lambda_{cx}}$)
  - `<secondary_lambdacy_mean>` ($\overline{\lambda}_{cy}$), `<secondary_lambdacy_stdev>` ($\sigma_{\lambda_{cy}}$)
  - `<secondary_gamma_mean>` ($\overline{\gamma}$) and `<secondary_gamma_stdev>` ($\sigma_\gamma$)

so that the parameters $\lambda_{cx}$, $\lambda_{cy}$ and $\gamma$ in (52) are realized such that

$$\begin{aligned}\lambda_{cx} &= \overline{\lambda}_{cx} + \tilde{x}_1 \sigma_{\lambda_{cx}} \\ \lambda_{cy} &= \overline{\lambda}_{cy} + \tilde{x}_2 \sigma_{\lambda_{cy}} \\ \gamma &= \overline{\gamma} + \tilde{x}_3 \sigma_{\gamma}\end{aligned}$$

where $\tilde{x}_1,\ \tilde{x}_2,\ \tilde{x}_2 \sim U\,[0,1]$.

7.4.5. *Tertiary Branch Parameters.* Unlike the stem, primary or the secondary branches, the tertiary branches are assumed to be randomly distributed throughout the tree crown disconnected from the tree structure described so far. The structure of tertiary branches are not as complex as the other types of branches either, consequently fewer parameters are required to control tertiary branch realizations.

- `Number of Tertiaries`: Since **PolSARproSim**$^{+}$relies on a hybrid-stochastic scattering model for sub-wavelength scale vegetation (see Section 6.4.2), not all tertiary branches are realized. Instead a representative sample are imaged and the scattered field is scaled accordingly. The number of tertiaries is estimated based on the size of the crown and four parameters:
  - `<tertiary_volume_fraction>`: The fraction of the crown volume to be occupied by tertiary branches, $t_{vf}$.
  - `<tertiary_number_factor>`: Factor to scale the number of tertiary branches, $t_{nf}$
  - `<tertiary_minimum_length>`: Minimum length of the tertiary branches (in meters) $t_{l_{min}}$
  - `<tertiary_minimum_radius>`: Minimum radius of the tertiary branches (in meters) $t_{r_{min}}$

The number of tertiaries realized, $N_T$, is then calculated as

$$N_T = \min\left\{n_t, \left\lfloor \frac{t_{nf} c_{area}}{\delta_x \delta_y} \right\rfloor\right\}$$

where the canopy area $c_{area} = \pi c_{radius}^2$ for the maximum canopy radius $c_{radius}$, $\delta_x$ and $\delta_y$ are the azimuth and range resolutions, while $n_t$ is calculated as

$$n_t = \left\lfloor t_{vf} \frac{c_{volume}}{t_{bv}} \right\rfloor$$

where $c_{volume}$ is the crown volume (computed based on the shape of the crown), and $t_{bv}$ the tertiary branch volume is given by

$$t_{bv} = \pi t_r^2 t_l$$

for tertiary branch radius, $t_r$ calculated as

$$t_r = \max\left\{t_{r_{min}}, \frac{\bar{s}_r^2}{\bar{p}_r}\right\}$$

where $\bar{p}_r$ and $\bar{s}_r$ are the mean radii of the primary and secondary branches respectively, while the tertiary branch length, $t_l$ is calculated using

$$t_l = \max\left\{t_{l_{min}}, \frac{\bar{s}_l^2}{\bar{p}_l}\right\}$$

where $\bar{p}_l$ and $\bar{s}_l$ are the mean lengths of the primary and secondary branches respectively.

- `<tertiary_moisture>`: This double precision parameter between 0 and 0.7 is the gravimetric moisture content ($M_g$) of the tertiary branches used to derive their electrical properties, i.e. their effective permittivities, which are estimated using (58) as given in [13].

7.4.6. *Foliage Parameters.* Much like tertiary branches, leaves are also imaged using Monte-Carlo sampling methods, consequently the parameters needed to control foliage parameters are similar as well. They are

- `<leaf_shape>`: this character string can either be 'broadleaf' or 'needle' imitating the deciduous and coniferous leaf shapes
- `Leaf Size`: The size of leaves can be controlled using six parameters, the mean and standard deviation of the width, height and thickness that are specified using the six tags
  - `<leaf_width_mean>`: the mean value of the leaf width in meters, $\bar{l}_w$
  - `<leaf_width_stdev>`: the standard deviation of the leaf width in meters $\sigma_{lw}$
  - `<leaf_height_mean>`: the mean value of the leaf height in meters, $\bar{l}_h$
  - `<leaf_height_stdev>`: the standard deviation of the leaf height in meters $\sigma_{lh}$
  - `<leaf_thickness_mean>`: the mean value of the leaf thickness in meters, $\bar{l}_t$
  - `<leaf_thickness_stdev>`: the standard deviation of the leaf thickness in meters $\sigma_{lt}$

  the leaf dimensions are then realized such that

$$l_w = \bar{l}_w + (\tilde{x}_1 - 0.5)\,\sigma_{lw}$$
$$l_h = \bar{l}_h + (\tilde{x}_2 - 0.5)\,\sigma_{lh}$$
$$l_t = \bar{l}_t + (\tilde{x}_3 - 0.5)\,\sigma_{lt}$$

  where $l_w$, $l_h$ and $l_t$ are the realized width, height and thickness of a leaf while $\tilde{x}_1,\ \tilde{x}_2,\ \tilde{x}_3 \sim U\,[0,1]$.
- `Leaf Number`: Similar to the number of tertiaries, the number of leaves realized is calculated based on the canopy parameters, and controlled by two parameters specified under the tags
  - `<leaf_volume_fraction>`: The fraction of the canopy volume occupied by leaves, $f_{vf}$, as a double precision number between 0 and 1.
  - `<leaf_number_factor>`: Scale factor for the number of leaves, $f_n$.

The number of leaves is then calculated such that

$$N_l = \left\lfloor \max\left\{ n_f, \frac{f_n c_{area}}{\delta_x \delta_y} \right\} \right\rfloor$$

where the canopy area $c_{area} = \pi c_{radius}^2$ for the maximum canopy radius $c_{radius}$, $\delta_x$ and $\delta_y$ are the azimuth and range resolutions, while $n_f$ is calculated as

$$n_f = f_{vf} \frac{c_{volume}}{l_{vol}}$$

where $c_{volume}$ is the crown volume (computed based on the shape of the crown), and $l_{vol}$ the leaf volume is given by

$$l_{vol} = \begin{cases} \pi l_w l_h^2 & \text{if species is 'needle'} \\ l_w l_h l_t & \text{if species is 'broadleaf'} \end{cases}$$

- `<leaf_moisture>`: This double precision parameter between 0 and 0.7 is the gravimetric moisture content ($M_g$) of the leaves used to derive their electrical properties, i.e. their effective permittivities, which are estimated using (58) as given in [13].

### 7.5. **Graphic File Format.**

Following the input of parameters and construction of the forest scene a small graphic image can be calculated and output as a guide to the visual interpretation of the resulting SAR images. Graphic image files have the extension 'gri'. Procedures for reading and writing the graphic image files are provided as part of **PolSARproSim**$^+$. This library is self-contained and defined in the files 'GraphicIMage.h' and 'GraphicIMage.c'. Graphics files with the extension '.gri' are binary files with the following format

| | |
|---|---|
| long | np, the total number of pixels in the image |
| int | nx, the number of pixels in the 'x'-direction |
| int | ny, the number of pixels in the 'y'-direction |
| int | n, the number of char bytes in the filename |
| char×n | the filename |
| int | Ninfo, the number of bytes of ancillary text information |
| char× Ninfo | the ancillary text information |
| graphic_pixel×n | the image data |

where the graphic pixel is defined as follows

```
typedef struct graphic_pixel_tab {
  unsigned char   red;
  unsigned char   green;
  unsigned char   blue;
} graphic_pixel;
```

# Appendix A. Finding Ray Intersections

## A.1. General.

In all cases we define a ray with a start position $\vec{p}$, direction $\hat{a}$, and point of intersection $\vec{s} = \vec{p} + \alpha\hat{a}$.

## A.2. Intersection with a Plane.

Let the plane have normal unit vector $\hat{n}_p$ and pass through the point $\vec{p}_0$. Let the unit vectors $\hat{x}_p$ and $\hat{y}_p$, lying in the plane, form an orthonormal set with $hatn_p$. The point of intersection is a solution of

$$\vec{s} = \vec{p} + \alpha\hat{a} + A\hat{x}_p + B\hat{y}_p \tag{167}$$

From which it follows that if then $|\hat{a} \cdot \hat{n}_p| > 0$ then

$$\alpha = \frac{(\vec{p}_0 - \vec{p}) \cdot \hat{n}_p}{\hat{a} \cdot \hat{n}_p} \tag{168}$$

otherwise there is no intersection. Note that it is possible to find $\alpha < 0$ and the success of the algorithm depends upon the sign of $\alpha$ when looking for branch lengths.

## A.3. Intersection with a Cylinder.

Let the cylinder have axial unit vector $\hat{z}_c$ passing through the base point $\vec{c}_0$. Let the unit vectors $\hat{x}_c$ and $\hat{y}_c$, form an orthonormal set with $\hat{z}_c$. The point of intersection is a solution of

$$\begin{aligned} \vec{s} &= \vec{p} + \alpha\hat{a} \\ &= \vec{c}_0 + z_s\hat{z}_c + R\left(\cos\theta_s\hat{x}_c + \sin\theta_s\hat{y}_c\right) \end{aligned} \tag{169}$$

It follows that

$$\hat{z}_c \cdot \vec{s} = (\hat{z}_c \cdot \vec{p}) + \alpha\left(\hat{z}_c \cdot \hat{a}\right) = (\hat{z}_c \cdot \vec{c}_0) + z_s \tag{170}$$

$$\hat{x}_c \cdot \vec{s} = (\hat{x}_c \cdot \vec{p}) + \alpha\left(\hat{x}_c \cdot \hat{a}\right) = (\hat{x}_c \cdot \vec{c}_0) + R\cos\theta_s \tag{171}$$

$$\hat{y}_c \cdot \vec{s} = (\hat{y}_c \cdot \vec{p}) + \alpha\left(\hat{y}_c \cdot \hat{a}\right) = (\hat{y}_c \cdot \vec{c}_0) + R\sin\theta_s \tag{172}$$

We need only use the last two equations to recover an expression for $\alpha$ as the solution to a quadratic equation:

$$\left[\hat{x}_c \cdot (\vec{p} - \vec{c}_0) + \alpha\left(\hat{x}_c \cdot \hat{a}\right)\right]^2 + \left[\hat{y}_c \cdot (\vec{p} - \vec{c}_0) + \alpha\left(\hat{y}_c \cdot \hat{a}\right)\right]^2 = R^2 \tag{173}$$

writing

$$A = \hat{x}_c \cdot (\vec{p} - \vec{c}_0), \quad B = \hat{x}_c \cdot \hat{a}, \quad C = \hat{y}_c \cdot (\vec{p} - \vec{c}_0), \quad D = \hat{y}_c \cdot \hat{a} \tag{174}$$

then

$$\alpha^2\left(B^2+D^2\right)+2\alpha\left(AB+CD\right)+\left(A^2+C^2-R^2\right)=0 \tag{175}$$

so that

$$\alpha=\frac{-\left(AB+CD\right)\pm\sqrt{\left(AB+CD\right)^2-\left(B^2+D^2\right)\left(A^2+C^2-R^2\right)}}{\left(B^2+D^2\right)} \tag{176}$$

Note that if the argument to the square root is negative the ray does not intersect the cylinder. Also if both B and D are zero then the ray direction is parallel with the cylinder axis and we have either no intersection or an infinity of intersections if the ray lies in the cylinder surface. Otherwise there are two solutions, and the choice of solution will depend upon the application.

### A.4. **Intersection with a Cone.**

Let the cone have axial unit vector $\hat{z}_c$ passing through the base point $\vec{c}_0$ to the tip of the cone at $t_c=\vec{c}_0+h\hat{z}_c$. Let the unit vectors $\hat{x}_c$ and $\hat{y}_c$, form an orthonormal set with $\hat{z}_c$. The point of intersection is a solution of

$$\begin{aligned}\vec{s}&=\vec{p}+\alpha\hat{a}\\&=\vec{c}_0+z_s\hat{z}_c+R\left(z_s\right)\left(\cos\theta_s\hat{x}_c+\sin\theta_s\hat{y}_c\right)\end{aligned} \tag{177}$$

where $0\leq z_s\leq h$ and $R\left(z_s\right)=\left(h-z_s\right)\tan\beta$. It follows that

$$\hat{z}_c\cdot\vec{s}=\left(\hat{z}_c\cdot\vec{p}\right)+\alpha\left(\hat{z}_c\cdot\hat{a}\right)=\left(\hat{z}_c\cdot\vec{c}_0\right)+z_s \tag{178}$$

$$\hat{x}_c\cdot\vec{s}=\left(\hat{x}_c\cdot\vec{p}\right)+\alpha\left(\hat{x}_c\cdot\hat{a}\right)=\left(\hat{x}_c\cdot\vec{c}_0\right)+R\left(z_s\right)\cos\theta_s \tag{179}$$

$$\hat{y}_c\cdot\vec{s}=\left(\hat{y}_c\cdot\vec{p}\right)+\alpha\left(\hat{y}_c\cdot\hat{a}\right)=\left(\hat{y}_c\cdot\vec{c}_0\right)+R\left(z_s\right)\sin\theta_s \tag{180}$$

We need to recover an expression for $\alpha$ as the solution to a quadratic equation

$$\left[\hat{x}_c\cdot\left(\vec{p}-\vec{c}_0\right)+\alpha\left(\hat{x}_c\cdot\hat{a}\right)\right]^2+\left[\hat{y}_c\cdot\left(\vec{p}-\vec{c}_0\right)+\alpha\left(\hat{y}_c\cdot\hat{a}\right)\right]^2=R^2\left(z_s\right) \tag{181}$$

with

$$R^2\left(z_s\right)=\left(h-z_s\right)^2\tan^2\beta \tag{182}$$

$$z_s=\hat{z}_c\cdot\left(\vec{p}-\vec{c}_0\right)+\alpha\left(\hat{z}_c\cdot\hat{a}\right) \tag{183}$$

writing

$$A=\hat{x}_c\cdot\left(\vec{p}-\vec{c}_0\right),\quad B=\hat{x}_c\cdot\hat{a},\quad C=\hat{y}_c\cdot\left(\vec{p}-\vec{c}_0\right), \tag{184}$$

$$D=\hat{y}_c\cdot\hat{a},\quad E=\hat{z}_c\cdot\left(\vec{p}-\vec{c}_0\right),\quad F=\hat{z}_c\cdot\hat{a} \tag{185}$$

we have

$$(A+\alpha B)^2+(C+\alpha D)^2=(h-E-\alpha F)^2\tan^2\beta \tag{186}$$

which is quadratic in $\alpha$ and has the solution

$$\alpha=\frac{-b\pm\sqrt{b^2-4ac}}{2a} \tag{187}$$

where

$$a=B^2+D^2-F^2\tan^2\beta \tag{188}$$

$$b=2\left(AB+CD+F\left(h-E\right)\tan^2\beta\right) \tag{189}$$

$$c=A^2+C^2-\left(h-E\right)^2\tan^2\beta \tag{190}$$

The same caveats apply: if the denominator is zero, or if the argument to the square root is negative then there is no solution, otherwise two solutions will exist and these should not lie beyond the tip of the cone.

### A.5. **Intersection with a Spheroid.**

Let the spheroid have axial unit vector $\hat{z}_s$ passing through the base point $\vec{c}_0$ to the tip of the cone at $t_s=\vec{c}_0+h\hat{z}_s$. Note that the spheroid has semi-axes $a_1$ (major, $\hat{z}_s$) direction and $a_2$ (minor) and that $0\le h\le 2a_1$. Let the unit vectors $\hat{x}_s$ and $\hat{y}_s$, form an orthonormal set with $\hat{z}_s$. The point of intersection is a solution of

$$\begin{aligned}\vec{s}&=\vec{p}+\alpha\hat{a}\\&=\vec{c}_0+z_s\hat{z}_s+R\left(z_s\right)\left(\cos\theta_s\hat{x}_s+\sin\theta_s\hat{y}_s\right)\end{aligned} \tag{191}$$

where $0\le z_s\le h$. Note that now we have for the spheroid radius

$$R^2\left(z_s\right)=\tan^2\beta\left[a_1^2-\left(a_1+h+z_s\right)^2\right],\quad \tan\beta=a_2/a_1. \tag{192}$$

Following our previous analysis

$$\hat{z}_s\cdot\vec{s}=\left(\hat{z}_s\cdot\vec{p}\right)+\alpha\left(\hat{z}_s\cdot\hat{a}\right)=\left(\hat{z}_s\cdot\vec{c}_0\right)+z_s \tag{193}$$

$$\hat{x}_s\cdot\vec{s}=\left(\hat{x}_s\cdot\vec{p}\right)+\alpha\left(\hat{x}_s\cdot\hat{a}\right)=\left(\hat{x}_s\cdot\vec{c}_0\right)+R\left(z_s\right)\cos\theta_s \tag{194}$$

$$\hat{y}_s\cdot\vec{s}=\left(\hat{y}_s\cdot\vec{p}\right)+\alpha\left(\hat{y}_s\cdot\hat{a}\right)=\left(\hat{y}_s\cdot\vec{c}_0\right)+R\left(z_s\right)\sin\theta_s \tag{195}$$

We need to recover an expression for as the solution to a quadratic equation

$$\left[\hat{x}_s\cdot\left(\vec{p}-\vec{c}_0\right)+\alpha\left(\hat{x}_s\cdot\hat{a}\right)\right]^2+\left[\hat{y}_s\cdot\left(\vec{p}-\vec{c}_0\right)+\alpha\left(\hat{y}_s\cdot\hat{a}\right)\right]^2=R^2\left(z_s\right) \tag{196}$$

and $z_s=\hat{z}_s\cdot\left(\vec{p}-\vec{c}_0\right)+\alpha\left(\hat{z}_s\cdot\hat{a}\right)$. Writing

$$A = \hat{x}_s \cdot (\vec{p} - \vec{c}_0)\,, \quad B = \hat{x}_s \cdot \hat{a}, \quad C = \hat{y}_s \cdot (\vec{p} - \vec{c}_0)\,, \tag{197}$$

$$D = \hat{y}_s \cdot \hat{a}, \quad E = \hat{z}_s \cdot (\vec{p} - \vec{c}_0)\,, \quad F = \hat{z}_s \cdot \hat{a} \tag{198}$$

we can rewrite (196) to be quadratic in $\alpha$ with solution

$$\alpha = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \tag{199}$$

where

$$a = B^2 + D^2 + F^2 \tan^2 \beta \tag{200}$$

$$b = 2\left(AB + CD + F\left(a_1 - h + E\right)\tan^2 \beta\right) \tag{201}$$

$$c = A^2 + C^2 - \left(a_1^2 - \left(a_1 - h + E\right)^2\right)\tan^2 \beta \tag{202}$$

The same caveats apply: if the denominator is zero, or if the argument to the square root is negative then there is no solution, otherwise two solutions will exist and these should not lie beyond the tip of the spheroid.

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